

Affine Branching and Modular Properties of the $\text{PSU}(1,1|2)/\text{SU}(1|2)$ WZW Coset Model

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We study the affine branching functions and modular properties of the WZW coset $\text{PSU}(1,1|2)/\text{SU}(1|2)$. Starting from the finite-dimensional branching rules, we derive the affine character decompositions for typical representations with finite-dimensional ground states and extend them to representations whose ground states belong to the discrete and principal continuous series of the non-compact $\mathfrak{sl}(2, \mathbb{R})$ factor. Finite-ground-state atypical representations are reconstructed from their discrete-series branches, including the exceptional vacuum case. We derive the modular transformations of the typical $\widehat{\mathfrak{psu}}(1,1|2)_k$ characters with the spectral-flow labels kept explicit and examine the modular closure of the typical part of the auxiliary bulk state space. Irreducible atypical numerator and denominator characters are treated through formal resolutions into character families on which the corresponding modular kernels act. Finally, the modular transformation of the coset branching functions is obtained from a generalized intertwining relation by embedding the discrete sums over the allowed denominator charges as distributions in the continuously labelled Kac-character basis.

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1. Introduction

Conformal field theories based on Lie supergroups arise in several areas of two-dimensional quantum field theory. The supergroup $\text{PSU}(1,1|2)$ is particularly familiar from string theory on $\text{AdS}_3 \times S^3$, where $\text{PSU}(1,1|2)_L \times \text{PSU}(1,1|2)_R$ forms the relevant superisometry group [PR99, BVW99]. The representation theory of the corresponding affine algebra, including its discrete and principal continuous representations and their spectral-flow images, has therefore been studied in detail [GQS07]. In the hybrid formulation this supergroup description has also been used explicitly to determine the massless and first massive string spectra on $\text{AdS}_3 \times S^3$ [GG11, Ger12]. More recently, the exceptional $\widehat{\mathfrak{psu}}(1,1|2)_1$ WZW theory has received renewed attention in the tensionless $\text{AdS}_3 \times S^3 \times T^4$ context, with subsequent work developing free-field realizations and the treatment of spectrally flowed states [EGG19, GNS22, DKN24]. This special level is distinct from the generic representation-theoretic regime considered below. The coset considered in the present work,

$$\frac{\text{PSU}(1,1|2)}{\text{SU}(1|2)},$$

is, however, not the supercoset describing the $\text{AdS}_3 \times S^3$ background. Here it is considered primarily as a non-compact supergroup coset CFT, motivated by the role of related superspace theories in two-dimensional critical systems [Ber97, SS07, CS11, QS13].

The corresponding compact coset $\text{PSU}(2|2)/\text{SU}(1|2)$ realizes the projective superspace $\mathbb{CP}^{1|2}$, and its GKO construction belongs to the class of superspace parafermion theories discussed in Ref. [CS11]. These theories generalize ordinary parafermionic cosets and admit exactly marginal deformations. More generally, supergroup and supercoset CFTs occur in descriptions of critical and disordered systems, where non-unitarity, indecomposable representations and continuously varying scaling dimensions can occur [Ber97, SS07, QS13]. The model studied here is the corresponding non-compact, hyperbolic counterpart of the compact $\mathbb{CP}^{1|2}$ theory. As discussed in Sec. 2, its bosonic body is

$$\frac{\text{SU}(1,1)}{\text{U}(1)} \cong \mathbb{H}^2$$

instead of $\mathbb{CP}^1 \cong S^2$.

The two WZW theories entering the coset have special conformal properties. The $\text{PSU}(1,1|2)$ WZW model has central charge $c = -2$, while the $\text{SU}(1|2)$ WZW model has $c = 0$ [GQS07, SS07]. Related supergroup and supercoset models possess exactly marginal deformations. Along such conformal families the central charge remains fixed, while individual scaling dimensions may vary continuously with the deformation parameter [MQS08, CS11, CMS13]. In the compact superspace-parafermion construction, the existence of the marginal perturbation follows from the special representation-theoretic properties of the coset rather than simply from the value of the central charge [CS11].

The conformal data of numerator and denominator enter directly into the branching functions. As shown explicitly in Sec. 8, if a numerator representation Λ branches into a denominator representation λ , the leading power of the corresponding branching function has the schematic form

$$\text{ord}_q \Psi_\lambda^{(\Lambda)} = m_\Lambda - m_\lambda + r_\lambda^\Lambda.$$

Here the modular anomalies m contain the conformal weights and central-charge contributions of numerator and denominator, while the non-negative integer r_λ^Λ gives the lowest affine grade at which the corresponding denominator representation occurs. The branching functions therefore encode not only representation-theoretic multiplicities but also the graded conformal data of the coset theory.

Characters, physical representation spaces and modular properties of the two ingredient theories are known to a considerable extent [GQS07, SS07, STT05]. For $\widehat{\mathfrak{psu}}(1,1|2)_k$, the physically relevant spectrum involves discrete and principal continuous representations and their spectral-flow images, together with atypical representations and the indecomposable structures built from them. The denominator theory $\widehat{\mathfrak{su}}(1|2)_k$ likewise contains typical and atypical representations with qualitatively different modular behaviour. To obtain the corresponding coset theory one has to determine how the numerator characters decompose into denominator characters and how the resulting branching functions transform under the modular group. These two problems are linked because the branching decomposition must intertwine the modular transformations of numerator and denominator.

The present work develops and extends calculations begun in the author's diploma thesis [Moz12], where the affine branching was obtained for typical representations with finite-dimensional ground states. Here this analysis is extended to representations with discrete and principal continuous infinite-dimensional ground states and to finite-ground-state atypical representations, and is supplemented by the modular analysis of the constituent characters and the resulting coset branching functions.

Secs. 3 and 4 collect the finite-dimensional representation theory and branching relations that underlie the affine construction, while Secs. 5 and 6 introduce the affine characters with finite- and infinite-dimensional ground states. The fixed-charge decompositions of the negative-level $\widehat{\mathfrak{sl}}(2)$ characters needed for the left diagonal branching are derived in Sec. 7. For typical numerator representations with finite-dimensional ground states, Sec. 8 factorizes the affine branching into a left diagonal branching involving finite-Weyl negative-level $\widehat{\mathfrak{sl}}(2)$ characters and a compact $\widehat{\mathfrak{su}}(2)$ diagonal branching. Sec. 9 extends the left branching to representations whose ground states belong to the discrete and principal continuous series of the non-compact $\mathfrak{sl}(2, \mathbb{R})$ factor, replacing the finite-Weyl characters by the corresponding infinite-dimensional $\widehat{\mathfrak{sl}}(2)$ characters. For finite-ground-state atypical numerator representations, whose branching decomposition contains irreducible atypical denominator characters explicitly, Sec. 10 separates these contributions from the complementary typical denominator characters and treats the vacuum separately. In each case the affine decomposition refines the horizontal branching by keeping track of the grades at which the denominator representations occur.

The modular analysis begins with the spectral-flow structure in Sec. 11. In Sec. 12 the modular transformations of the typical $\widehat{\mathfrak{psu}}(1,1|2)_k$ characters are derived with the spectral-flow label retained explicitly, rather than being absorbed into unfolded representation labels as is possible for the discrete-series representations. This keeps the complete spectral-flow orbit generated under the S -transformation visible and provides a common notation for both the discrete and principal continuous series. Sec. 13 then examines modular closure of the corresponding typical part of the auxiliary bulk state space. The irreducible atypical denominator and numerator characters are treated in Secs. 14 and 15 through formal resolutions into character families on which the modular kernels act naturally.

Finally, Sec. 16 derives the modular transformation of the coset branching functions directly from compatibility of the full character decomposition with the numerator and denominator modular transformations. The discrete sums over the allowed denominator charges are embedded as distributions in the continuously labelled Kac-character basis on which the denominator Fourier-sine kernel acts. This distributional embedding is equivalent to the original discrete branching decomposition and introduces no additional branching data; inserting its definition and performing the charge integral reproduces the original discrete sum. Irreducible atypical denominator contributions are first resolved into the same Kac-character family, allowing typical and resolved atypical branching distributions to be treated within one modular intertwining framework.

The resulting character decompositions and modular transformations provide the character and modular data needed for a modular description of the non-compact $\text{PSU}(1,1|2)/\text{SU}(1|2)$ coset. They do not by themselves constitute a construction of the complete interacting modular-invariant bulk theory. In particular, the full treatment of the indecomposable atypical bulk sector lies beyond the scope of the present work. The results obtained here should instead be viewed as the character-theoretic ingredients required for such a construction.

2. Preliminaries

Background space-time $\text{AdS}_3 \times S^3 \times \mathcal{M}_4$. The string backgrounds motivating the representation theory considered below are of the form $\text{AdS}_3 \times S^3 \times \mathcal{M}_4$, with $\mathcal{M}_4 = T^4$ or $K3$ [PR99, arXiv, Sec. 1]. The AdS_3 factor provides the setting for the $\text{AdS}_3/\text{CFT}_2$ correspondence. Its $\mathfrak{sl}(2, \mathbb{R})_{\text{L}} \oplus \mathfrak{sl}(2, \mathbb{R})_{\text{R}} \cong \mathfrak{so}(2, 2)$ isometry algebra is identified with the global conformal algebra of the two-dimensional boundary theory [GKS98, arXiv, Sec. 2.2]. The S^3 factor has isometry group $\text{SO}(4)$, locally $\text{SU}(2)_{\text{L}} \times \text{SU}(2)_{\text{R}}$. Together with the AdS_3 isometries, these compact factors form the bosonic subalgebra of the $\text{PSU}(1,1|2)_{\text{L}} \times \text{PSU}(1,1|2)_{\text{R}}$ superisometry algebra and act non-trivially on the corresponding supercharges [PR99, arXiv, Sec. 3]. The internal factor $\mathcal{M}_4$ will not enter the character calculations below; the relevant structure for the present work is the $\text{PSU}(1,1|2)$ symmetry associated with the $\text{AdS}_3 \times S^3$ sector.

$\text{AdS}_3 \times S^3$ as a homogeneous space. The connected component of the isometry group of AdS_3 is

$$\text{SO}_0(2,2) \cong \frac{\text{SL}(2, \mathbb{R})_{\text{L}} \times \text{SL}(2, \mathbb{R})_{\text{R}}}{\mathbb{Z}_2},$$

so we see that AdS_3 is described by a quadratic form of signature (2,2) invariant under $\text{SO}(2,2)$. The reason for the $\mathbb{Z}_2$ quotient is that, under the group-manifold identification of AdS_3 with $\text{SL}(2, \mathbb{R})$, the element $(-\mathbb{1}_2, -\mathbb{1}_2) \in \text{SL}(2, \mathbb{R})_{\text{L}} \times \text{SL}(2, \mathbb{R})_{\text{R}}$ acts trivially by left-right multiplication and must therefore be factored out. For the Lie-algebraic considerations below, and after passing to the universal cover, this discrete quotient is not relevant, so we suppress it. $\text{SL}(2, \mathbb{R})$ is the non-compact group of matrices with unit determinant and it is isomorphic to $\text{SU}(1,1)$. Its projective form $\text{PSL}(2, \mathbb{R}) \cong \text{PSU}(1,1)$ is the orientation-preserving isometry group of the hyperbolic plane $\mathbb{H}^2$ (Poincaré disk). For AdS_3 one uses the product $\text{SL}(2, \mathbb{R})_{\text{L}} \times \text{SL}(2, \mathbb{R})_{\text{R}}$ (or $\text{SU}(1,1)_{\text{L}} \times \text{SU}(1,1)_{\text{R}}$). In the supersymmetric extension the relevant real Lie superalgebra (see the paragraph on complexification below) is $\mathfrak{psu}(1,1|2)$, which is a real form of the complex Lie superalgebra $\mathfrak{psl}(2|2, \mathbb{C})$ [GQS07, arXiv, Sec. 2.1]. To unwrap the circular time-like direction

in AdS_3 , we pass to the universal cover $\widetilde{\text{SU}}(1,1)$. In the following, we drop the tilde and implicitly assume we work with the universal cover whenever global issues are relevant, since the Lie algebra structure is unchanged. The isomorphism between $\text{SL}(2, \mathbb{R})$ and $\text{SU}(1,1)$ lifts to their universal covers.

The S^3 part as the compact counterpart of AdS_3 can be described analogously, with $\text{SO}(4) \cong (\text{SU}(2)_L \times \text{SU}(2)_R)/\mathbb{Z}_2$. Its universal cover is $\text{SU}(2)_L \times \text{SU}(2)_R$. Again, since we will be interested in algebras only, the $\mathbb{Z}_2$ quotient will not play a role below. Up to the discrete identifications and coverings discussed above, the bosonic isometry group of $\text{AdS}_3 \times S^3$ is thus given by $\text{SU}(1,1)_L \times \text{SU}(1,1)_R \times \text{SU}(2)_L \times \text{SU}(2)_R$. The pairs $\text{SU}(1,1)_L \times \text{SU}(2)_L$ and $\text{SU}(1,1)_R \times \text{SU}(2)_R$ are the bosonic subgroups of two corresponding $\text{PSU}(1,1|2)$ factors. Adjoining the fermionic generators completes them to the superisometry group $\text{PSU}(1,1|2)_L \times \text{PSU}(1,1|2)_R$.

The stabilizer of AdS_3 is given by $\text{SO}_0(2, 1)$, and that of S^3 is $\text{SO}(3)$ [PR99, arXiv, Sec. 3]. We have the isomorphisms $\text{SO}_0(2, 1) \cong \text{SU}(1,1)_{\text{diag}}/\mathbb{Z}_2$ and $\text{SO}(3) \cong \text{SU}(2)_{\text{diag}}/\mathbb{Z}_2$. The supersymmetric extension is therefore described by the supercoset

$$\frac{\text{PSU}(1,1|2)_L \times \text{PSU}(1,1|2)_R}{(\text{SU}(1,1) \times \text{SU}(2))_{\text{diag}}},$$

whose bosonic body is $\text{AdS}_3 \times S^3$. Geometrically, the fermionic generators in the numerator do not belong to the stabilizer of the reference point, but generate fermionic coset directions; the denominator group is therefore purely bosonic [PR99, arXiv, Sec. 3].

Algebras and complexification. Complexification enters in two ways. Geometrically, AdS_3 and S^3 can be viewed as different real slices of the same complexified quadric. On the group side, the complex algebra $\mathfrak{sl}(2, \mathbb{C})$ has distinct real forms: $\mathfrak{su}(2)$ for the compact factors and $\mathfrak{sl}(2, \mathbb{R}) \cong \mathfrak{su}(1,1)$ for the non-compact factors. In representation theory it is convenient to work with the complexified algebra (e.g. $\mathfrak{sl}(2, \mathbb{C})$ or $\mathfrak{psl}(2|2, \mathbb{C})$), since highest-weight methods and character formulas are naturally formulated there, while the choice of real form determines the admissible physical representations. In physics, however, the isometries are real groups: for each of the left- and right-moving factors of $\text{AdS}_3 \times S^3$ the relevant real supergroup is $\text{PSU}(1,1|2)$, not its complexification. Concretely, among the real forms of the complex algebra $\mathfrak{psl}(2|2, \mathbb{C})$ are [FSS96, Sec. 31 and Table 16]

- $\mathfrak{psu}(2|2)$, compact, bosonic part: $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$,
- $\mathfrak{psu}(1,1|2)$, non-compact, bosonic part: $\mathfrak{su}(1,1) \oplus \mathfrak{su}(2)$.

Thus the relevant real form for each of the left- and right-moving factors of $\text{AdS}_3 \times S^3$ is $\mathfrak{psu}(1,1|2)$, a real form of $\mathfrak{psl}(2|2, \mathbb{C})$ [GQS07, arXiv, Sec. 2.1].

The algebra $\mathfrak{su}(1|2)$. As in the case of $\mathfrak{psu}(1,1|2)$, one starts from the complex Lie superalgebra $\mathfrak{sl}(1|2, \mathbb{C})$ and chooses an appropriate real form. The compact real form $\mathfrak{su}(1|2)$ has bosonic subalgebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2)$. In the coset considered in this work it appears as the denominator algebra. The isomorphic complex algebra $\mathfrak{sl}(2|1, \mathbb{C})$ is frequently used in the literature [SS07]; throughout the present work we use the notation $\mathfrak{sl}(1|2, \mathbb{C})$ in order to match the notation $\mathfrak{su}(1|2)$ for the corresponding real form.

Coset construction of $\mathbb{CP}^{1|2}$. The complex projective superspace $\mathbb{CP}^{1|2}$ is the space of even lines through the origin in $\mathbb{C}^{2|2}$ [CMQ⁺10, Sec. 2.1]. The group $\text{SL}(2|2, \mathbb{C})$ acts transitively on these lines. Its central scalar transformations leave every projective line fixed, so the same transitive action is described effectively by $\text{PSL}(2|2, \mathbb{C})$. If P denotes the full stabilizer of a chosen even line, then

$$\mathbb{CP}^{1|2} \cong \frac{\text{PSL}(2|2, \mathbb{C})}{P}. \quad (1)$$

The subgroup $\text{SL}(1|2, \mathbb{C})$ can be embedded such that it preserves the chosen line. It is nevertheless smaller than the full stabilizer P : after choosing a decomposition $\mathbb{C}^{2|2} = L \oplus W$, $W \cong \mathbb{C}^{1|2}$, with L the chosen even line, the stabilizer also contains transformations that add components along L to vectors in W . These transformations change the chosen complement W without changing the line L itself. They are part of the full stabilizer but are absent from the block-diagonal $\text{SL}(1|2, \mathbb{C})$ subgroup. Consequently,

$$\frac{\text{PSL}(2|2, \mathbb{C})}{\text{SL}(1|2, \mathbb{C})} \longrightarrow \mathbb{CP}^{1|2} \quad (2)$$

is a homogeneous fibration with fibre $P/\text{SL}(1|2, \mathbb{C}) \cong \mathbb{C}^{1|2}$. Correspondingly, $\text{PSL}(2|2, \mathbb{C})/\text{SL}(1|2, \mathbb{C})$ has complex dimension $(2|4)$, whereas $\mathbb{CP}^{1|2}$ has complex dimension $(1|2)$. This is the supersymmetric analogue of the familiar bosonic distinction between $\text{SL}(2, \mathbb{C})/\mathbb{C}^\times$ and the holomorphic projective space $\mathbb{CP}^1 \cong \text{SL}(2, \mathbb{C})/B$, where $\mathbb{C}^\times$ denotes the multiplicative group of non-zero complex numbers and B is the full stabilizer of a line. The important distinction from the unitary real forms below is that preserving L alone does not select or preserve a particular complement W .

The relation to $\mathbb{CP}^{1|2}$ is seen most directly from the compact real form. Up to global central identifications, the compact coset $\text{PSU}(2|2)/\text{SU}(1|2)$ realizes $\mathbb{CP}^{1|2}$. Equivalently, its standard unitary presentation is [CMQ⁺10, Eq. (2.3)]

$$\mathbb{CP}^{1|2} \cong \frac{\text{U}(2|2)}{\text{U}(1) \times \text{U}(1|2)}.$$

Indeed, $\text{U}(2|2)$ acts transitively on the even lines in $\mathbb{C}^{2|2}$, while the stabilizer of a chosen line consists of $\text{U}(1)$ acting on that line and $\text{U}(1|2)$ acting on its orthogonal complement. The unitary condition implies that a transformation preserving the line also preserves its orthogonal complement, so that the additional shear directions present in the complex parabolic stabilizer are absent. Passing to the special and projective groups removes the corresponding central factors and gives the compact presentation above, up to the stated global identifications. Its bosonic body is therefore

$$\frac{\text{U}(2)}{\text{U}(1) \times \text{U}(1)} \cong \mathbb{CP}^1 \cong \frac{\text{SU}(2)}{\text{U}(1)} \cong S^2. \quad (3)$$

The model considered in this work is the corresponding non-compact counterpart, $\text{PSU}(1,1|2)/\text{SU}(1|2)$. The same stabilizer argument applies to this unitary real form: the invariant Hermitian form forces the stabilizer of the chosen even line to preserve its orthogonal complement, so no additional shear directions occur. At the level of the bosonic body, the compact factor $\text{SU}(2)/\text{U}(1) \cong S^2$ is replaced by

$$\frac{\text{SU}(1,1)}{\text{U}(1)} \cong \mathbb{H}^2, \quad (4)$$

while the four odd directions remain. Thus the model may be viewed geometrically as the hyperbolic, non-compact counterpart of the compact $\mathbb{CP}^{1|2}$ model, rather than as $\mathbb{CP}^{1|2}$ itself.

The quotient notation is used in two related but distinct senses in this work. Geometrically, G/H denotes a homogeneous superspace that may serve as the target space of a sigma model [CMQ⁺10]. In the GKO construction, by contrast, the coset is defined at the level of affine current algebras: the denominator theory is factored out from the numerator theory, with the coset stress tensor and central charge obtained by subtracting the corresponding denominator contributions [GKO85]. For the models considered here, the relation between these two notions of coset emerges in a large- k limit, where k is the level of the affine current algebra. The marginally perturbed GKO coset is expected to approach the sigma model on the corresponding homogeneous superspace as $k \rightarrow \infty$ [CS11, Sec. IV]. In the standard WZW normalization the target-space metric scales with k , so the curvature in world-sheet units decreases as $1/k$ and the sigma model becomes semiclassical.

For completeness, the compact $\text{PSU}(2|2)/\text{SU}(1|2)$ model can be placed in the broader classification of Ref. [CS11]. One of the families considered there is of complex super-Grassmannian type,

$$\frac{\widehat{\mathfrak{psu}}(n+m|n+m)_k}{\widehat{\mathfrak{su}}(n-1|n)_k \oplus \widehat{\mathfrak{su}}(m+1|m)_k},$$

where the integers m and n determine the ranks of the supergroups. Here, “super-Grassmannian” refers to the supersymmetric analogue of an ordinary Grassmannian; in particular, $\mathbb{CP}^{1|2}$ is the projective case parametrizing even lines of $\mathbb{C}^{2|2}$. For $m = n = 1$, the first denominator factor is trivial and the family reduces to the compact $\text{PSU}(2|2)/\text{SU}(2|1)$ coset, equivalently $\text{PSU}(2|2)/\text{SU}(1|2)$ in the convention used here.

The same reference also considers the separate $\text{PSU}(2n|2n)/\text{OSP}(2n|2n)$ family. For $n = 1$ the two families coincide at the level of affine current algebras, since

$$\widehat{\mathfrak{osp}}(2|2)_{2k} \cong \widehat{\mathfrak{su}}(2|1)_k.$$

Correspondingly, the compact projective superspace also admits the realization

$$\frac{\text{PSU}(2|2)}{\text{OSP}(2|2)} \cong \mathbb{CP}^{1|2}$$

[CCMS10, Eq. (1.7)]. Thus, at the affine-algebra level, the $\text{OSP}(2|2)$ formulation does not give an independent coset in this special case, but places the compact $\mathbb{CP}^{1|2}$ model simultaneously in the two families considered there. This alternative realization will not enter the character calculations below.

Character formulas. Highest-weight modules are generated by acting with lowering operators on a highest-weight state. At special weights, descendants may themselves be annihilated by all raising operators; such *singular vectors* generate proper submodules. Irreducible highest-weight representations are obtained by quotienting the corresponding maximal submodules, and their characters may be described by standard highest-weight character formulas [Hum72, Secs. 20.2–20.3 and Sec. 24.3]. As the simplest example, for $\mathfrak{su}(2)$ the ordinary bosonic character of the

spin- j representation is

$$\chi_j(z) = \sum_{m=-j}^j z^m = z^{-j} + z^{-j+1} + \dots + z^j, \quad j \in \frac{\mathbb{N}_0}{2}. \quad (5)$$

Throughout, $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For the non-compact real form $\mathfrak{su}(1,1) \cong \mathfrak{sl}(2, \mathbb{R})$, the unitary series relevant here are infinite-dimensional and comprise the discrete, principal, and complementary series. We briefly summarize their zero-mode characters, since the discrete and principal series will reappear in the affine theory [GQS07, arXiv, Sec. 2.2]. For the discrete series we use the labels $(\pm, j)$, with $j < 0$. Their zero-mode characters are [GQS07, arXiv, Eq. (2.10)]

$$\chi_{(\pm, j)}(z) = \frac{z^{\mp j}}{1 - z^{\pm 1}} = \sum_{n \geq 0} z^{\mp j \pm n}.$$

The $+$ and $-$ signs denote the lowest- and highest-weight branches, respectively, and correspond to the two opposite Laurent expansions. The principal series is labelled by (j, α) , with $j \in -\frac{1}{2} + i\mathbb{R}$, $0 \leq \alpha < 1$. Its weight character is the bilateral distribution [GQS07, arXiv, Eq. (2.11)]

$$\chi_{(j, \alpha)}(z) = \sum_{n \in \mathbb{Z}} z^{\alpha + n}. \quad (6)$$

Thus the weight multiplicities are independent of j , while j fixes the Casimir $-j(j+1)$. The bilateral sum is understood distributionally rather than as an ordinarily convergent Laurent series: writing $z = e^{2\pi i x}$, it is the Dirac-comb distribution

$$\sum_{r \in \mathbb{Z}} z^{\alpha + r} = e^{2\pi i \alpha x} \sum_{n \in \mathbb{Z}} \delta(x - n),$$

and is therefore supported at $z = 1$ on the Cartan circle. A complementary series also exists but will not enter the calculations below.

For the Lie superalgebras considered below, one similarly starts from a representation of the even subalgebra and acts with the odd lowering generators. Their fermionic nature produces an exterior-algebra factor in the resulting Kac module. For generic values of the representation labels the Kac module is irreducible and is called typical. At special values it becomes reducible; we refer to such reducible Kac modules as atypical Kac modules. The corresponding irreducible atypical modules occur as composition factors and must be distinguished from the reducible Kac modules themselves. The explicit atypicality conditions and character relations needed for $\mathfrak{sl}(1|2)$ and $\mathfrak{psl}(2|2)$ will be given in the following sections [SS07, GQS07].

Upon affinization, the representations discussed above form the ground-state spaces of modules. Acting with negative current modes generates additional affine grades, so that the characters acquire a q -dependence in addition to the Cartan variables. Singular vectors and reducibility persist in the affine setting and may occur at higher affine grades. The explicit affine characters and the conventions needed for them will be introduced below.

Conventions relative to the diploma thesis. We use slightly different conventions from those of the author's unpublished diploma thesis [Moz12], on which part of the notation and the representation-theoretic background of this work are based. Throughout the present paper, z_1

refers to the Cartan variable of the first, non-compact factor and, after restriction to $\mathfrak{su}(1|2)$, to its $\mathfrak{u}(1)$ charge, whereas z_2 refers to the Cartan variable of the compact $\mathfrak{su}(2)$ factor. Accordingly, numerator representations are labelled as $[j_1, j_2]$, with j_1 referring to the first (non-compact) and j_2 to the second (compact) factor. For $\mathfrak{su}(1|2)$ we use the shifted and reordered labels $\{b, j\}$ rather than $\{j, b\}$ used in the thesis, so that atypicality occurs at $b = \pm j$. Moreover, χ denotes the sR supercharacter in the present work; consequently the fermionic factors carry the corresponding supertrace signs (and, in the affine formulas, θ_1 occurs rather than the theta function used in the ordinary-character convention of the thesis). Formulas taken from the thesis are understood with these convention changes.

3. Finite-dimensional representations of $\mathfrak{su}(1|2)$

We denote finite-dimensional Kac modules of $\mathfrak{su}(1|2)$ by $\{b, j\}$, using the shifted and reordered convention fixed in Sec. 2, with $j \in \frac{\mathbb{N}}{2}$ and $b \in \mathbb{C}$. For $b \neq \pm j$ the Kac module $\{b, j\}$ is irreducible and typical, whereas at $b = \pm j$ it becomes reducible and indecomposable [SS07, arXiv, Sec. 2.1.2]. Its sR supercharacter is

$$\chi_{\{b, j\}}(z_1, z_2) = z_1^{b - \frac{1}{2}} \chi_f^{\text{sR}}(z_1, z_2) \chi_{j - \frac{1}{2}}(z_2), \quad (7)$$

with the fermionic supercharacter in the super-Ramond (sR) sector

$$\chi_f^{\text{sR}}(z_1, z_2) = (1 - z_1^{1/2} z_2^{1/2})(1 - z_1^{1/2} z_2^{-1/2}) = 1 - (z_2^{1/2} + z_2^{-1/2}) z_1^{1/2} + z_1. \quad (8)$$

The polynomial χ_f^{sR} arises universally from the two odd generators of $\mathfrak{su}(1|2)$, which transform in the $\mathfrak{su}(2)$ doublet $(\frac{1}{2})_{1/2}$; equivalently, they have $\mathfrak{su}(2)$ -weights $\pm \frac{1}{2}$ and b -charge $+\frac{1}{2}$. More precisely, χ_f^{sR} is the supercharacter of the exterior (Grassmann) algebra generated by these fermions, encoding the alternating sum of the vacuum, the two single-fermion states, and their product. Acting with these fermions shifts the $\mathfrak{su}(2)$ weights by $\pm \frac{1}{2}$, which leads to the factorization into a bosonic part $z_1^{b-1/2} \chi_{j-1/2}(z_2)$ and a universal fermionic contribution χ_f^{sR} .

Here sR denotes the Ramond-sector supertrace convention; at the finite-dimensional level only the corresponding supertrace signs are relevant. We restrict to this convention throughout the following. In accordance with the conventions fixed in Sec. 2, z_1 denotes the $\mathfrak{u}(1)$ fugacity, whereas z_2 denotes the Cartan fugacity of $\mathfrak{su}(2)$.

For a $\mathbb{Z}_2$ -graded module M , ΠM denotes the parity-reversed module. We use the shorthand $-M := \Pi M$, so that at the level of supercharacters $\chi_{-M} = -\chi_M$. The $\mathbb{Z}_2$ -graded decomposition under the even subalgebra $\mathfrak{su}(1|2)_0 = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ is

$$\begin{aligned} \{b, j\}|_{\mathfrak{su}(1|2)_0} &= (j - \tfrac{1}{2})_{b-1/2} \oplus -(j)_b \oplus -(j-1)_b \oplus (j - \tfrac{1}{2})_{b+1/2} \\ &= V_f \otimes (j - \tfrac{1}{2})_{b-1/2}, \end{aligned}$$

with $V_f = (0)_0 \oplus -(\frac{1}{2})_{1/2} \oplus (0)_1$, and for $j = \frac{1}{2}$ it is

$$\{b, \tfrac{1}{2}\}|_{\mathfrak{su}(1|2)_0} = (0)_{b-1/2} \oplus -(\tfrac{1}{2})_b \oplus (0)_{b+1/2}.$$

With this convention, the minus signs in these decompositions keep track of the $\mathbb{Z}_2$ grading and give the corresponding signs in the sR supercharacter. At the atypical values $b = \pm j$, the Kac modules $\{\pm j, j\}$ are reducible and indecomposable. We denote the corresponding irreducible atypical modules by $\{j\}_\pm$; their $\mathbb{Z}_2$ -graded decomposition is

$$\{j\}_\pm|_{\mathfrak{su}(1|2)_0} = -(j)_{\pm j} \oplus (j - \frac{1}{2})_{\pm(j+1/2)},$$

and for $j = 0$ the two atypical branches coincide, and we define

$$\{0\} := \{0\}_+ = \{0\}_- = -(0)_0. \quad (9)$$

The overall parity convention for an irreducible atypical module is conventional: a parity reversal multiplies its supercharacter by -1 . The chosen convention places the minus sign on the component generated by a single fermionic action.

For $j \geq \frac{1}{2}$, the indecomposable Kac modules $\{\pm j, j\}$ have the composition structure of Ref. [SS07, arXiv, Eq. (2.5)], which in our parity convention can be written as the short exact sequence

$$0 \longrightarrow \{j\}_\pm \longrightarrow \{\pm j, j\} \longrightarrow -\{j - \frac{1}{2}\}_\pm \longrightarrow 0. \quad (10)$$

At the level of supercharacters, the short exact sequence gives [SS07, arXiv, Eq. (2.7)]

$$\chi_{\{\pm j, j\}} = \chi_{\{j\}_\pm} - \chi_{\{j - \frac{1}{2}\}_\pm}. \quad (11)$$

Iterating this relation yields the formal expansion of the irreducible atypical supercharacter in terms of Kac-module supercharacters [SS07, arXiv, Eq. (2.8)],

$$\chi_{\{j\}_\pm} = - \sum_{n=1}^{\infty} \chi_{\{\pm(j+\frac{n}{2}), j+\frac{n}{2}\}}. \quad (12)$$

The same complex Lie superalgebra also admits the non-compact real form $\mathfrak{su}(1|1,1)$ (usually denoted $\mathfrak{su}(1,1|1)$ in the convention of Ref. [BCM05]), whose even subalgebra is $\mathfrak{u}(1) \oplus \mathfrak{su}(1,1)$. This real form will not play a direct role below; the denominator algebra of the coset considered here remains $\mathfrak{su}(1|2)$.

4. Finite-dimensional representations of $\mathfrak{psu}(2|2)$ and decompositions

The sR supercharacters of the finite-dimensional Kac modules $[j_1, j_2]$, with $j_1, j_2 \in \frac{\mathbb{N}_0}{2}$, are [GQS07, arXiv, Eq. (2.6)]

$$\chi_{[j_1, j_2]}(z_1, z_2) = \chi_F(z_1, z_2) \chi_{j_1}(z_1) \chi_{j_2}(z_2).$$

The four odd generators entering the Kac construction transform in the bifundamental representation $(\frac{1}{2}, \frac{1}{2})$ of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and generate the corresponding exterior algebra. In the case of $\mathfrak{sl}(1|2)$ we have seen that the two corresponding odd generators generate two singlets (the identity and $F^- \bar{F}^-$, in the notation of Ref. [Moz12]) and one doublet obtained by a single fermionic

action on the highest-weight state. In the case of $\mathfrak{psu}(2|2)$ there are four such odd generators, whose exterior algebra generates a 16-dimensional supermultiplet,

$$V_F \equiv [0, 0] = 2(0, 0) \oplus -2(\tfrac{1}{2}, \tfrac{1}{2}) \oplus (1, 0) \oplus (0, 1).$$

Accordingly, the fermionic supercharacter in the sR sector can be reorganized into $\chi_F(z_1, z_2) = (\chi_{1/2}(z_1) - \chi_{1/2}(z_2))^2$, or, with $\chi_f(z_1, z_2) = z_1^{1/2}(\chi_{1/2}(z_1) - \chi_{1/2}(z_2))$, equivalently into [GQS07, arXiv, Eqs. (2.7), (2.8)]

$$\chi_F(z_1, z_2) = z_1^{-1} \chi_f^2(z_1, z_2) = \chi_f(z_1, z_2) \chi_f(z_1^{-1}, z_2).$$

This factorization reflects the structure of the odd part of $\mathfrak{psu}(2|2)$. The fermionic generators span the representation $(\tfrac{1}{2}, \tfrac{1}{2})$ of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. Upon restricting the first $\mathfrak{su}(2)$ to its Cartan subalgebra $\mathfrak{u}(1)$ this representation decomposes into two charge sectors,

$$(\tfrac{1}{2}, \tfrac{1}{2}) \longrightarrow (\tfrac{1}{2})_{+1/2} \oplus (\tfrac{1}{2})_{-1/2}.$$

Equivalently, the four odd generators split into two pairs with opposite $\mathfrak{u}(1)$ charges. The exterior algebra of the odd space therefore factorizes into the tensor product of the exterior algebras generated by these two pairs, which explains the factorized form of χ_F . The $\mathbb{Z}_2$ -graded decomposition under the even subalgebra $\mathfrak{psu}(2|2)_{\bar{0}} = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ is

$$[j_1, j_2] \Big|_{\mathfrak{psu}(2|2)_{\bar{0}}} = V_F \otimes (j_1, j_2).$$

Using $V_F = (0)_{-1} \otimes V_f^2$, we obtain at the level of sR supercharacters the parity-refined form of the branching rule of Ref. [GQS05, Eq. (3.15)],

$$\chi_{[j_1, j_2]} = \sum_{b=-j_1}^{j_1} (\chi_{\{b+\frac{1}{2}, j_2+\frac{1}{2}\}} - \chi_{\{b, j_2+1\}} - \chi_{\{b, j_2\}} + \chi_{\{b-\frac{1}{2}, j_2+\frac{1}{2}\}}). \quad (13)$$

Here and in the following, sums over b run in steps of one. For $j_2 = 0$ the formal term $\{b, 0\}$ is absent, and Eq. (13) reduces to

$$\chi_{[j_1, 0]} = \sum_{b=-j_1}^{j_1} (\chi_{\{b+\frac{1}{2}, \frac{1}{2}\}} - \chi_{\{b, 1\}} + \chi_{\{b-\frac{1}{2}, \frac{1}{2}\}}).$$

The outer sum over b arises from restricting the first $\mathfrak{su}(2)$ factor to its Cartan $\mathfrak{u}(1)$, while the four terms inside the brackets reflect the exterior algebra of the two fermionic directions complementary to the embedded $\mathfrak{su}(1|2)$. Their exterior degrees are 0, 1, 1, 2, respectively, which explains the parity pattern $+, -, -, +$.

At the module level, the interpretation changes when one of the $\mathfrak{su}(1|2)$ Kac modules appearing in Eq. (13) reaches an atypical wall. In the present branching this occurs when one of the allowed b -weights satisfies

$$b = \pm j_2 \quad \text{or} \quad b = \pm(j_2 + 1).$$

At such values the corresponding Kac module becomes reducible and indecomposable, with the corresponding submodule structure described in Sec. 3. Since b runs through the weights $-j_1, -j_1 + 1, \dots, j_1$, such reducible Kac summands can occur only when the corresponding weight lattice reaches these walls, in particular when $j_1 \geq j_2$. In such cases the character identity remains valid, while the corresponding module restriction may involve non-trivial indecomposable or projective modules [GQS05, Eq. (3.16)].

Geometrically, the compact supercoset $\mathbb{CP}^{1|2}$ has bosonic body $\mathbb{CP}^1 \cong \text{SU}(2)/\text{U}(1)$, described by the homogeneous fibration

$$\text{U}(1) \longrightarrow \text{SU}(2) \longrightarrow \text{SU}(2)/\text{U}(1) \cong \mathbb{CP}^1.$$

The same reduction $\text{SU}(2) \rightarrow \text{U}(1)$ appears in the branching of the characters: a spin- j_1 representation decomposes into the weights $b = -j_1, -j_1 + 1, \dots, j_1$, so that b plays the same role as the magnetic quantum number in the usual angular-momentum decomposition. The two additional odd directions of $\mathbb{CP}^{1|2}$ form a fermionic doublet under the remaining $\text{SU}(2)$; their four Grassmann states account for the four terms in the branching rule. Thus the b -decomposition reflects the bosonic $\mathbb{CP}^1$ part, while the remaining structure of the branching reflects the two odd directions of the supercoset.

Reducible atypical finite-dimensional Kac modules of $\mathfrak{psu}(2|2)$ occur for $j_1 = j_2$ [GQS07, arXiv, Eq. (2.12)]. We denote the layered structure of the corresponding indecomposable Kac modules by arrows between successive composition layers. With the parity assignments made explicit in our sR convention, the composition structures of Ref. [GQS07, arXiv, Eqs. (2.13),(2.15),(2.16)] read as follows. For $j \geq 1$ one has

$$[j, j] : \quad [j] \longrightarrow -[j + \tfrac{1}{2}] \oplus -[j - \tfrac{1}{2}] \longrightarrow [j].$$

For $j = \frac{1}{2}$ the structure becomes

$$[\tfrac{1}{2}, \tfrac{1}{2}] : \quad [\tfrac{1}{2}] \longrightarrow -[1] \longrightarrow -[0] \oplus -[0] \longrightarrow [\tfrac{1}{2}].$$

Finally, for $j = 0$ one obtains

$$[0, 0] : \quad [0] \longrightarrow -[\tfrac{1}{2}] \longrightarrow [0].$$

Iterating the relations implied by the composition structure gives the following formal supercharacter expansion:

$$\chi_{[j]} = - \sum_{n=1}^{\infty} n \chi_{[j+\frac{n}{2}, j+\frac{n}{2}]}, \quad j \geq \tfrac{1}{2}. \quad (14)$$

For $j = 0$, either of the two discrete-series branches may be used, but the sign has to be chosen consistently throughout the expansion. Thus, for fixed $s \in \{+, -\}$, Eq. (2.32) of Ref. [GQS07, arXiv] can be written, after shifting the summation index by one, as

$$\chi_{[0]} = \sum_{n=1}^{\infty} n \chi_{[(s, -\frac{n+1}{2}), \frac{n-1}{2}]}. \quad (15)$$

Here $[0]$ denotes the one-dimensional trivial representation.

Translating the module restriction formula of Ref. [GQS05, Eq. (3.14)] to our parity convention gives, for $j \geq \frac{1}{2}$,

$$[j] = -\{j\}_+ \oplus -\{j\}_- \oplus \bigoplus_{b=-j+\frac{1}{2}}^{j-\frac{1}{2}} \{b, j + \frac{1}{2}\}. \quad (16)$$

For $j = 0$ the same restriction reduces to

$$[0] = -\{0\}.$$

5. Affine characters with finite-dimensional ground states

For $\ell \in \mathbb{N}$ and $m \in \mathbb{Z}$ we define

$$\Theta_m^{(\ell)}(z, q) := \sum_{t \in \mathbb{Z}} q^{(m+2\ell t)^2/(4\ell)} z^{(m+2\ell t)/2}, \quad (17)$$

and the corresponding extended $\hat{\mathfrak{u}}(1)_\ell$ character, for $j \in \frac{\mathbb{Z}}{2}$, by

$$K_j^{(\ell)}(z, q) := \frac{\Theta_{2j}^{(\ell)}(z, q)}{\eta(q)}.$$

The labels are periodic, $\Theta_{m+2\ell}^{(\ell)} = \Theta_m^{(\ell)}$, $K_{j+\ell}^{(\ell)} = K_j^{(\ell)}$. For a non-zero level ℓ we furthermore use the fixed-charge quantities

$$\bar{\Theta}_m^{(\ell)}(z, q) := q^{m^2/(4\ell)} z^{m/2}, \quad \bar{K}_j^{(\ell)}(z, q) := \frac{\bar{\Theta}_{2j}^{(\ell)}(z, q)}{\eta(q)} = \frac{q^{j^2/\ell} z^j}{\eta(q)}. \quad (18)$$

Here the barred quantities may also be used for continuous or complex charge labels. Thus, $\Theta^{(\ell)}$ contains the full affine charge-lattice sum, while $\bar{\Theta}^{(\ell)}$ retains a single fixed-charge term. Correspondingly, we refer to $K^{(\ell)}$ as the extended $\hat{\mathfrak{u}}(1)_\ell$ character and to $\bar{K}^{(\ell)}$ as the fixed-charge Heisenberg Fock character. For negative ℓ the Gaussian exponent changes sign, so that the full charge-lattice sum does not converge for $|q| < 1$. For the abelian algebra $\hat{\mathfrak{u}}(1)_\ell$ one has $g^\vee = 0$, so no dual-Coxeter shift of the level occurs.

We denote the universal affine fermionic factor by

$$F(z_1, z_2, q) := \frac{\theta_1(z_1^{\frac{1}{2}} z_2^{\frac{1}{2}}, q) \theta_1(z_1^{-\frac{1}{2}} z_2^{\frac{1}{2}}, q)}{\eta^2(q)}.$$

The affine Kac supercharacters for $\widehat{\mathfrak{su}}(1|2)_k$ in the sR sector are [SS07, arXiv, Secs. 3.1–3.2, Eq. (3.2)]

$$\chi_{\{b, j\}}^{(k)}(z_1, z_2, q) = F(z_1, z_2, q) \bar{K}_b^{(-k-1)}(z_1, q) \chi_{j-\frac{1}{2}}^{(k-1)}(z_2, q), \quad (19)$$

where $k \in \mathbb{N}$, $j = \frac{1}{2}, 1, \dots, \frac{k}{2}$, and $b \in \mathbb{C}$. Away from the affine atypicality loci $b = \pm j + t(k+1)$, $t \in \mathbb{Z}$, the corresponding Kac modules are irreducible and typical; on these loci they are reducible

atypical Kac modules. Using $\chi_i^{(1)} = K_i^{(1)}$ for $i = 0, \frac{1}{2}$, the fermionic part can be written as

$$\begin{aligned} F(z_1, z_2, q) &= - \sum_{i=0, \frac{1}{2}} (-1)^{2i} K_i^{(1)}(z_1, q) K_{|i-\frac{1}{2}|}^{(1)}(z_2, q) \\ &= - \frac{1}{\eta^2(q)} \sum_{i=0, \frac{1}{2}} (-1)^{2i} \Theta_{2i}^{(1)}(z_1, q) \Theta_{|2i-1|}^{(1)}(z_2, q). \end{aligned} \quad (20)$$

Structurally, $\widehat{\mathfrak{sl}}(1|2)_k$ is the centrally extended loop algebra of $\mathfrak{sl}(1|2)$. The odd generators give rise to four fermionic currents, whose modes form the odd part of the affine algebra. In the realization used to derive the character formula above [SS07, arXiv, Sec. 3.1], the character factorizes into two fermionic pairs and commuting bosonic factors. Each fermionic pair contributes a factor θ_1/η , while the bosonic factors give the integrable $\widehat{\mathfrak{sl}}(2)_{k-1}$ character and the fixed-charge Heisenberg factor $\bar{K}_b^{(-k-1)}$.

As an example, consider the affine module at level k with ground-state representation $G = \{\frac{k+2}{2}, \frac{k}{2}\}$. Its ground states have conformal weight $h = -1$, and the next affine grade is generated by the current modes. At the level of supercharacters one therefore finds

$$\chi_G^{(k)} = q^{-1} \chi_G - \chi_{\{0,1\}} \chi_G - \chi_{\{\frac{k+2}{2}, \frac{k+2}{2}\}} + O(q).$$

Here,

$$\{0, 1\} = (\frac{1}{2})_{-1/2} \oplus -(1)_0 \oplus -(0)_0 \oplus (\frac{1}{2})_{1/2}$$

denotes the parity-reversed adjoint representation, so that the minus sign in the grade-one contribution restores the ordinary adjoint parity. For this special ground-state representation the grade-one contribution in the corresponding affine Verma module, viewed as a horizontal $\mathfrak{sl}(1|2)$ module, is non-semisimple and contains a finite-dimensional projective cover. Within this grade-one contribution the bosonic affine singular vectors generate the Kac submodule $\{\frac{k+2}{2}, \frac{k+2}{2}\}$. This submodule is quotiented out of the affine Verma module, which accounts for its negative contribution in the character expansion [SS07, arXiv, Sec. 3.3]. The detailed structure of the projective cover will not be needed below. This example illustrates how non-semisimple horizontal structure and affine singular vectors are encoded in the affine supercharacter.

The irreducible atypical supercharacters for $\widehat{\mathfrak{su}}(1|2)_k$ are [SS07, arXiv, Sec. 3.3, Eq. (3.6)]

$$\chi_{\{j\}_{\pm}}^{(k)}(z_1, z_2, q) = \frac{iF(z_1, z_2, q)}{\eta(q)\theta_1(z_2, q)} \sum_{a=\pm 1} a \sum_{t \in \mathbb{Z}} q^{(2j+1)t+(k+1)t^2} \frac{z_1^{\pm(j+\frac{1}{2})} z_2^{a(j+\frac{1}{2}+t(k+1))}}{1 - z_1^{\pm\frac{1}{2}} z_2^{\frac{a}{2}} q^t}. \quad (21)$$

In terms of the fixed-charge theta factors introduced above, this may be written as

$$\chi_{\{j\}_{\pm}}^{(k)}(z_1, z_2, q) = \frac{iF(z_1, z_2, q)}{\eta(q)\theta_1(z_2, q)} \sum_{a=\pm 1} a \sum_{t \in \mathbb{Z}} \frac{\bar{\Theta}_{\pm(2j+1)}^{(-k-1)}(z_1, q) \bar{\Theta}_{a(2j+1+2t(k+1))}^{(k+1)}(z_2, q)}{1 - z_1^{\pm\frac{1}{2}} z_2^{\frac{a}{2}} q^t}. \quad (22)$$

For $j = \frac{1}{2}, 1, \dots, \frac{k}{2}$, these characters describe the unflowed finite-ground-state series whose ground states transform in the finite-dimensional atypical representations $\{j\}_{\pm}$. More generally, the labels $\{j\}_{\pm}$, $j \in \frac{\mathbb{N}}{2}$, enumerate the infinite family of irreducible affine atypical modules. Labels beyond the finite-ground-state range are affine atypical labels associated with the

spectral-flow orbit structure and should not be interpreted as the spins of additional unflowed finite-dimensional atypical ground states [SS07, arXiv, Secs. 3.3–3.4]. The vacuum $j = 0$ is exceptional; the two formal branches $\{0\}_+$ and $\{0\}_-$ give the same vacuum character.

Expanding the character to first order in q one finds

$$\chi_{\{j\}_{\pm}}^{(k)} = \chi_{\{j\}_{\pm}} - q \chi_{\{0,1\}} \chi_{\{j\}_{\pm}} + O(q^2).$$

For the vacuum, $\{0\}_+ = \{0\}_- = \{0\} = -(0)_0$, so that the additional parity reversal of the trivial ground state changes the sign once more:

$$\chi_{\{0\}}^{(k)} = \chi_{\{0\}} + q \chi_{\{0,1\}} + O(q^2).$$

The grade-one space has the expected ordinary dimension 8.

For later use, let $j_n \geq 0$ denote the non-negative highest-weight label in the finite-Weyl representations of $\widehat{\mathfrak{sl}}(2)_{-k-2}$. This notation will distinguish it from the discrete-series label $j_d < -\frac{1}{2}$ introduced below. We define the corresponding finite-Weyl character by [GQS07, arXiv, Eq. (3.37)]

$$\chi_{j_n}^{(-k-2)}(z, q) = -\frac{i q^{-\frac{(j_n+1/2)^2}{k}}}{\theta_1(z, q)} (z^{j_n+\frac{1}{2}} - z^{-j_n-\frac{1}{2}}). \quad (23)$$

In the $\widehat{\mathfrak{psu}}(2|2)_k$ character below, the first label j_1 plays the role of this finite-Weyl label. The affine Kac supercharacters with finite-dimensional ground states are [GQS07, arXiv, Eq. (3.36)]

$$\chi_{[j_1, j_2]}^{(k)}(z_1, z_2, q) = F^2(z_1, z_2, q) \chi_{j_1}^{(-k-2)}(z_1, q) \chi_{j_2}^{(k-2)}(z_2, q). \quad (24)$$

Here $k \geq 2$, $j_1 = 0, \frac{1}{2}, \dots, \frac{k+2}{2}$, and $j_2 = 0, \frac{1}{2}, \dots, \frac{k-2}{2}$. For $j_1 \neq j_2$ the corresponding Kac module is irreducible and typical, whereas for $j_1 = j_2$ it is a reducible atypical Kac module.

The corresponding irreducible atypical supercharacters for $j \geq \frac{1}{2}$ are [GQS07, arXiv, Eqs. (3.32), (3.39)]

$$\chi_{[j]}^{(k)}(z_1, z_2, q) = \frac{F^2(z_1, z_2, q)}{\theta_1(z_1, q) \theta_1(z_2, q)} \sum_{a, b=\pm 1} ab \sum_{t \in \mathbb{Z}} \frac{\bar{\Theta}_{2b(j+1)}^{(-k)}(z_1, q) \bar{\Theta}_{2a(j+1+t k)}^{(k)}(z_2, q)}{(1 - z_1^{\frac{b}{2}} z_2^{\frac{a}{2}} q^t)^2}, \quad (25)$$

where $j = \frac{1}{2}, 1, \dots, \frac{k-2}{2}$. In terms of the two discrete-series branches $[j]_{\pm}$ defined explicitly in Sec. 6, they satisfy the formal supercharacter identity

$$\chi_{[j]}^{(k)} = -\chi_{[j]_+}^{(k)} - \chi_{[j]_-}^{(k)}. \quad (26)$$

The vacuum case $j = 0$ is exceptional. Either of the two discrete-series branches gives the same vacuum character,

$$\chi_{[0]}^{(k)} = -\chi_{[-1/2]_+}^{(k)} = -\chi_{[-1/2]_-}^{(k)}. \quad (27)$$

The two expressions on the right are not independent contributions. Rather, they are different Laurent expansions of the same meromorphic character, corresponding to different expansion chambers. Thus, either branch represents the vacuum character by itself, and unlike the generic relation for $j \geq \frac{1}{2}$, the two branches are not to be added.

Expanding the vacuum character up to the first excited grade yields

$$\chi_{[0]}^{(k)} = q^{\frac{1}{12}} \chi_{[0]} + q^{\frac{13}{12}} \chi_{(1,0) \oplus (0,1) \oplus -2(\frac{1}{2}, \frac{1}{2})} \chi_{[0]} + O(q^{25/12}).$$

The non-trivial factor appearing at the first excited grade is in fact the adjoint atypical representation

$$[\frac{1}{2}] = (1, 0) \oplus (0, 1) \oplus -2(\frac{1}{2}, \frac{1}{2}).$$

Since $[0] = (0, 0) \equiv (0)_0 = -\{0\}_{\pm}$ and $[\frac{1}{2}] = -\{\frac{1}{2}\}_{+} \oplus -\{\frac{1}{2}\}_{-} \oplus \{0, 1\}$, the vacuum character may equivalently be written as

$$\chi_{[0]}^{(k)} = -q^{1/12} \chi_{\{0\}} + q^{13/12} (-\chi_{\{\frac{1}{2}\}_{+}} - \chi_{\{\frac{1}{2}\}_{-}} + \chi_{\{0,1\}}) + O(q^{25/12}). \quad (28)$$

The state space at the first excited grade therefore has ordinary dimension $3 + 3 + 8 = 14$.

6. Affine characters with infinite-dimensional ground states

For the physically relevant unflowed discrete series, i.e. the $w = 0$ spectral-flow sector (see Sec. 11), with $-\frac{k+1}{2} < j_1 < -\frac{1}{2}$ and $2j_2 = 0, \dots, k-2$, the characters of the typical (i.e. with $j_1 + j_2 + 1 \neq 0$) representations with discrete-series ground states read [GQS07, arXiv, Eqs. (3.27), (3.28)]

$$\chi_{[(\pm, j_1), j_2]}^{(k)}(z_1, z_2, q) = F^2(z_1, z_2, q) \chi_{(\pm, j_1)}^{(-k-2)}(z_1, q) \chi_{j_2}^{(k-2)}(z_2, q) \quad (29)$$

where

$$\chi_{(\pm, j_1)}^{(-k-2)}(z_1, q) = \frac{\pm i q^{-\frac{(j_1+1/2)^2}{k}}}{\theta_1(z_1, q)} z_1^{\mp(j_1+1/2)} = \frac{\pm i \bar{\Theta}_{\mp(2j_1+1)}^{(-k)}(z_1, q)}{\theta_1(z_1, q)}. \quad (30)$$

Formally continuing the two discrete-series branches to the finite-Weyl label $j_n \geq 0$ and comparing with Eq. (23) gives

$$\chi_{j_n}^{(-k-2)} = \chi_{(+, j_n)}^{(-k-2)} + \chi_{(-, j_n)}^{(-k-2)}. \quad (31)$$

Consequently, using Eqs. (24) and (29), the finite-ground-state $\widehat{\mathfrak{psu}}(1, 1|2)_k$ character satisfies the formal identity

$$\chi_{[j_1, j_2]}^{(k)} = \chi_{[(+, j_1), j_2]}^{(k)} + \chi_{[(-, j_1), j_2]}^{(k)}, \quad (32)$$

where the two characters on the right are understood by formal continuation of the discrete-series branches to the finite-Weyl label $j_1 \geq 0$. The atypicals with infinite-dimensional discrete-series ground states are denoted by $[j]_{\pm}$, with $j \geq 0$, and their supercharacters admit the formal branch resolutions [GQS07, arXiv, Eq. (3.32)]

$$\chi_{[j]_{\pm}}^{(k)}(z_1, z_2, q) = - \sum_{n=1}^{\infty} n \chi_{[(\pm, -(j+1+\frac{n}{2})), j+\frac{n}{2}]}^{(k)}(z_1, z_2, q). \quad (33)$$

This relation is understood as a formal identity of supercharacters, with each discrete-series branch taken in its corresponding Laurent expansion. The characters on the right-hand side are formally continued beyond the physical unflowed range when necessary. Its formal continuation to $j = -\frac{1}{2}$ yields the vacuum relation Eq. (27).

For $j_1 \in -\frac{1}{2} + i\mathbb{R}$, $0 \leq \alpha < 1$ ($\alpha \bmod 1$), and $j_2 = 0, \frac{1}{2}, \dots, \frac{k-2}{2}$, the characters of the principal continuous representations are given by [GQS07, arXiv, Eqs. (3.34), (3.35)]

$$\chi_{[(j_1, \alpha), j_2]}^{(k)}(z_1, z_2, q) = F^2(z_1, z_2, q) \chi_{(j_1, \alpha)}^{(-k-2)}(z_1, q) \chi_{j_2}^{(k-2)}(z_2, q) \quad (34)$$

where

$$\chi_{(j_1, \alpha)}^{(-k-2)}(z_1, q) = \frac{-iq^{-\frac{(j_1+1/2)^2}{k}}}{\theta_1(z_1, q)} (z^{1/2} - z^{-1/2}) \chi_{(j_1+\frac{1}{2}, \alpha)}(z_1). \quad (35)$$

Here the last factor is the bilateral principal-series weight distribution defined in Eq. (6), with $j \rightarrow j_1 + \frac{1}{2}$ and the same charge parameter α . All principal continuous representations in this parameter range are typical; no atypical principal-series representations occur [GQS07, arXiv, Sec. 2.2].

7. Negative-level $\widehat{\mathfrak{sl}}(2)$ string functions

After setting $j_1 = -\frac{1}{2}$ in Eq. (30) and subsequently taking $k \rightarrow 0$, the factor $-i/\theta_1(z_1, q)$ admits the following string-function decomposition into fixed-charge theta factors:

$$\frac{-i}{\theta_1(z_1, q)} = -s \chi_{(s, -\frac{1}{2})}^{(-2)}(z_1, q) = -s \sum_{\substack{m \in \mathbb{Z} \\ [m]_2=1}} \bar{c}_m^{((s, -\frac{1}{2}), -2)}(q) \bar{\Theta}_m^{(-2)}(z_1, q), \quad (36)$$

where $s = \pm 1$ labels the two discrete-series branches and

$$\begin{aligned} \bar{c}_m^{((s, -\frac{1}{2}), -2)}(q) &= -\frac{s}{\eta^3(q)} \sum_{\substack{l \in \mathbb{Z} \\ \text{sgn}(l+\frac{1}{2}) = \text{sgn}(-\frac{m}{2})}} (-1)^l \text{sgn}(l+\frac{1}{2}) q^{\frac{1}{2}(l-\frac{m}{2})^2} \\ &= -\frac{s}{\eta^3(q)} \sum_{l \in \mathbb{N}_0} (-1)^l q^{\frac{1}{2}(l+\frac{|m+1|}{2}+\frac{1}{2})^2}. \end{aligned} \quad (37)$$

Since $[m]_2 = 1$, the charge m is odd and hence nonzero. The sign condition $\text{sgn}(l+\frac{1}{2}) = \text{sgn}(-\frac{m}{2})$ restricts the contributing values of l to $l \geq 0$ or $l \leq -1$, respectively. The second expression rewrites both cases as a half-line sum over $l \in \mathbb{N}_0$. The exponent within this half-line sum can be reorganized as

$$\frac{1}{2} \left(l + \frac{|m+1|}{2} + \frac{1}{2} \right)^2 = \frac{1}{2} \left(l - \frac{m}{2} \right)^2 + (2l+1) \max \left(0, \frac{m+1}{2} \right).$$

This generalizes to arbitrary discrete series representations. In the following, $j_d < -\frac{1}{2}$ denotes the discrete-series label. For a given branch $s = \pm 1$, the allowed fixed-charge labels satisfy

$$m \in -2sj_d + 2\mathbb{Z}.$$

Define

$$B_s(j_d, m, l) = -k \left(\frac{l}{2} + \frac{s(2j_d+1)}{2k} \right)^2 + (k+2) \left(\frac{l}{2} - \frac{m}{2(k+2)} \right)^2$$

and

$$C_s(j_d, m) = \frac{m+1}{2} + s\left(j_d + \frac{1}{2}\right).$$

Then the discrete-series string functions may be written as

$$\bar{\bar{c}}_m^{((s, j_d), -k-2)}(q) = -\frac{s}{\eta^3(q)} \sum_{l \in \mathbb{N}_0} (-1)^l q^{B_s(j_d, m, l) + (2l+1) \max(0, C_s(j_d, m))}. \quad (38)$$

This is the half-line form of the discrete-series string function.

The level -2 , $j_d = -\frac{1}{2}$ string function in Eq. (37) is recovered by first setting $2j_d + 1 = 0$ and subsequently taking $k \rightarrow 0$, yielding

$$B_s\left(-\frac{1}{2}, m, l\right) \longrightarrow \frac{1}{2}\left(l - \frac{m}{2}\right)^2.$$

One then observes that

$$B_s(j_d, m, -l-1) - B_s(j_d, m, l) = (2l+1)C_s(j_d, m).$$

Hence, if $C_s(j_d, m) \leq 0$, then $\max(0, C_s(j_d, m)) = 0$ and the sum remains a sum over $l \geq 0$. If instead $C_s(j_d, m) > 0$, one has

$$B_s(j_d, m, l) + (2l+1)C_s(j_d, m) = B_s(j_d, m, -l-1),$$

so that the additional term involving the max-function can be absorbed by the reflection $l \mapsto -l-1$. In this way the half-line sum over $l \geq 0$ is converted into a half-line sum over $l \leq -1$. Since $(-1)^{-l-1} = -(-1)^l$, the reflected contribution acquires an additional minus sign. Both cases can therefore be combined into a single sum over all $l \in \mathbb{Z}$, with the relevant half-line selected by a sign condition. One thus obtains the second form of the discrete-series string function

$$\bar{\bar{c}}_m^{((s, j_d), -k-2)}(q) = -\frac{s}{\eta^3(q)} \sum_{\substack{l \in \mathbb{Z} \\ \text{sgn}(l + \frac{1}{2}) = \text{sgn}(-A_s(j_d, m))}} (-1)^l \text{sgn}(l + \frac{1}{2}) q^{B_s(j_d, m, l)}, \quad (39)$$

with

$$A_s(j_d, m) = \frac{m}{2} + s\left(j_d + \frac{1}{2}\right).$$

The charge condition $m \in -2sj_d + 2\mathbb{Z}$ is essential here. Indeed, writing $m = -2sj_d + 2r$ with $r \in \mathbb{Z}$, one finds $A_s(j_d, m) = r + \frac{s}{2} \in \mathbb{Z} + \frac{1}{2}$, so that $A_s(j_d, m)$ never vanishes. Since $C_s(j_d, m) = A_s(j_d, m) + \frac{1}{2}$, the sign of A_s distinguishes the cases $C_s \leq 0$ and $C_s > 0$ unambiguously and hence selects the relevant half-line of the l -sum.

For the finite-Weyl representations of $\widehat{\mathfrak{sl}}(2)_{-k-2}$, with non-negative highest-weight label $j_n \geq 0$ as introduced in Eq. (23), the corresponding finite-Weyl string function is obtained by formally continuing the two discrete-series string functions to $j_d = j_n$ and adding the resulting expressions,

$$\bar{\bar{c}}_m^{(2j_n, -k-2)}(q) = \bar{\bar{c}}_m^{((+, j_n), -k-2)}(q) + \bar{\bar{c}}_m^{((- , j_n), -k-2)}(q). \quad (40)$$

We denote the negative-level string functions for infinite-dimensional ground states by $\bar{\bar{c}}$ and those of the finite-Weyl representations by $\bar{c}$; the unbarred symbol c is reserved for the standard positive-level string functions. The quantities on the right-hand side of Eq. (40) are formal continuations

of the discrete-series string functions; their label $j_n \geq 0$ lies outside the discrete-series range. Equivalently,

$$\bar{c}_m^{(2j_n, -k-2)}(q) = -\frac{1}{\eta^3(q)} \sum_{l \in \mathbb{N}_0} (-1)^l [q^{E_+(j_n, m, l)} - q^{E_-(j_n, m, l)}], \quad (41)$$

where $E_s(j_n, m, l) = B_s(j_n, m, l) + (2l+1) \max(0, C_s(j_n, m))$. Using Eq. (39), each of the two formally continued discrete-series string functions may equivalently be written as a full integer sum with the corresponding sign restriction. Accordingly, the finite-Weyl character $\chi_{j_n}^{(-k-2)}$ admits the fixed-charge decomposition

$$\chi_{j_n}^{(-k-2)}(z, q) = \sum_{\substack{p \in \mathbb{Z} \\ p \equiv 2j_n \pmod{2}}} \bar{c}_p^{(2j_n, -k-2)}(q) \bar{\Theta}_p^{(-k-2)}(z, q). \quad (42)$$

As a consistency check, these finite-Weyl string-function expressions can be formally continued to positive integral level and, after restoring the affine Weyl-translation lattice, reorganized into the standard Kac-Peterson form of the $\widehat{\mathfrak{su}}(2)_k$ string functions [KP84]. Since the positive-level expression is not needed below, we do not reproduce this standard result here.

For the principal continuous series, the bilateral character in Eq. (35) leads to a different fixed-charge decomposition. In this case the factor $-i/\theta_1(z, q)$ cannot be decomposed separately using the discrete-series string functions above, because it occurs together with the Weyl numerator $z^{1/2} - z^{-1/2}$ and the bilateral weight distribution. The latter is distributionally supported at $z = 1$. Consequently, the regular combination $-i(z^{1/2} - z^{-1/2})/\theta_1(z, q)$ acts on this distribution through its value at $z = 1$, namely

$$\left. -i \frac{z^{1/2} - z^{-1/2}}{\theta_1(z, q)} \right|_{z=1} = \frac{1}{\eta^3(q)}.$$

The principal continuous character therefore admits the decomposition

$$\chi_{(j_1, \alpha)}^{(-k-2)}(z, q) = \sum_{p \in 2\alpha + 2\mathbb{Z}} \bar{c}_p^{((j_1, \alpha), -k-2)}(q) \bar{\Theta}_p^{(-k-2)}(z, q), \quad (43)$$

where

$$\bar{c}_p^{((j_1, \alpha), -k-2)}(q) = \frac{q^{-\frac{(j_1+1/2)^2}{k} + \frac{p^2}{4(k+2)}}}{\eta^3(q)}. \quad (44)$$

Here $p/2 \in \alpha + \mathbb{Z}$ is the Cartan charge. The p -dependence of the string function compensates the Gaussian factor contained in $\bar{\Theta}_p^{(-k-2)}$, so that the coefficient of the bare charge monomial $z^{p/2}$ is independent of p . The decomposition is understood distributionally in the Cartan variable.

8. Decomposition of typical characters with finite-dimensional ground states

After combining Eqs. (24) and (20), and factoring out one universal fermionic factor $F(z_1, z_2, q)$ required to reconstruct the $\widehat{\mathfrak{su}}(1|2)_k$ characters as given in Eq. (19), the remaining branching problem is to reorganize

$$-\frac{1}{\eta^2(q)} \sum_{i=0, \frac{1}{2}} (-1)^{2i} \Theta_{2i}^{(1)}(z_1, q) \Theta_{|2i-1|}^{(1)}(z_2, q) \chi_{j_1}^{(-k-2)}(z_1, q) \chi_{j_2}^{(k-2)}(z_2, q).$$

We refer to the decompositions in the z_1 and z_2 factors as the left and right diagonal branchings, respectively. On the right, this is the standard diagonal $\widehat{\mathfrak{su}}(2)$ branching $\widehat{\mathfrak{su}}(2)_{k-2} \oplus \widehat{\mathfrak{su}}(2)_1 \rightarrow \widehat{\mathfrak{su}}(2)_{k-1}$. For $i = 0, \frac{1}{2}$, the level-1 factor satisfies $K_i^{(1)} = \chi_i^{(1)}$, allowing the z_2 factor to be viewed as an $\widehat{\mathfrak{su}}(2)_1$ character.

Using the finite-Weyl string-function decomposition (42), we first perform the left diagonal branching. The left branching is associated with the diagonal Heisenberg embedding $\hat{\mathfrak{u}}(1)_{-k-1} \subset \widehat{\mathfrak{sl}}(2)_{-k-2} \oplus \hat{\mathfrak{u}}(1)_1$, and hence with the corresponding branching structure of

$$\frac{\widehat{\mathfrak{sl}}(2)_{-k-2} \oplus \hat{\mathfrak{u}}(1)_1}{\hat{\mathfrak{u}}(1)_{-k-1}}.$$

For the left diagonal branching, the relevant theta-product identity is

$$\begin{aligned} \Theta_{2i}^{(1)}(z_1, q) \bar{\Theta}_p^{(-k-2)}(z_1, q) &= \sum_{\substack{m \in \mathbb{Z} \\ m \equiv 2i+p \pmod{2}}} q^{\frac{[p(k+1)-m(k+2)]^2}{4(k+1)(k+2)}} \bar{\Theta}_m^{(-k-1)}(z_1, q) \\ &= \sum_{\substack{m \in \mathbb{Z} \\ m \equiv 2i+p \pmod{2}}} \bar{\Theta}_{p(k+1)-m(k+2)}^{((k+1)(k+2))}(1, q) \bar{\Theta}_m^{(-k-1)}(z_1, q). \end{aligned} \quad (45)$$

Since the string-function decomposition already sums over $p \equiv 2j_1 \pmod{2}$ (cf. Eq. (42)), the product condition $m \equiv 2i + p \pmod{2}$ implies $m \equiv 2(i + j_1) \pmod{2}$. For fixed m , the sum over p is therefore absorbed into the corresponding branching coefficient, while m labels the resulting diagonal $\hat{\mathfrak{u}}(1)_{-k-1}$ charge. We denote this coefficient by $\mathbb{L}_m^{(2j_1, -k-1)}(q)$. The left diagonal decomposition then reads

$$\frac{1}{\eta(q)} \Theta_{2i}^{(1)}(z_1, q) \chi_{j_1}^{(-k-2)}(z_1, q) = \sum_{\substack{m \in \mathbb{Z} \\ m \equiv 2(i+j_1) \pmod{2}}} \mathbb{L}_m^{(2j_1, -k-1)}(q) \bar{K}_{m/2}^{(-k-1)}(z_1, q), \quad (46)$$

where

$$\mathbb{L}_m^{(2j_1, -k-1)}(q) = \sum_{\substack{p \in \mathbb{Z} \\ p \equiv 2j_1 \pmod{2}}} \bar{c}_p^{(2j_1, -k-2)}(q) \bar{\Theta}_{p(k+1)-m(k+2)}^{((k+1)(k+2))}(1, q). \quad (47)$$

Here the superscript $(2j_1, -k-1)$ records the numerator label and the level of the resulting diagonal Heisenberg algebra; it should not be interpreted as the level of an additional affine representation carried by $\mathbb{L}$.

The left branching function can furthermore be reorganized in terms of formally continued finite-Weyl string functions at level $-k-1$. To keep the different charge indices distinct, let p_0 denote the original negative-level charge. With B_s denoting the quadratic exponent of the discrete-series string function introduced above, the complete exponent appearing in the convolution satisfies

$$\begin{aligned} B_s(j_1, p_0, l) &+ \frac{[p_0(k+1) - m(k+2)]^2}{4(k+1)(k+2)} \\ &= -k \left(\frac{l}{2} + \frac{s(2j_1+1)}{2k} \right)^2 + (k+1) \left(\frac{l}{2} - \frac{m}{2(k+1)} \right)^2 + \left(\frac{l}{2} + \frac{m-p_0}{2} \right)^2. \end{aligned} \quad (48)$$

For each of the two formally continued discrete-series string functions, introduce the bijective change of variables $p_0 = m + sr$. Since $p_0 \equiv 2j_1 \pmod{2}$, the new index satisfies $r \equiv 2j_1 - m \pmod{2}$. The first and third quadratic terms in Eq. (48) then combine according to

$$-k \left(\frac{l}{2} + \frac{s(2j_1 + 1)}{2k} \right)^2 + \left(\frac{l}{2} - \frac{sr}{2} \right)^2 = -(k-1) \left(\frac{l}{2} + \frac{s(2j_1 + r + 1)}{2(k-1)} \right)^2 + \frac{(2j_1 + 1 + kr)^2}{4k(k-1)}. \quad (49)$$

Together with the unchanged term $(k+1)(l/2 - m/[2(k+1)])^2$, the first term on the right-hand side reproduces precisely $B_s(j_1 + r/2, m, l)$ with k replaced by $k-1$, that is, the quadratic exponent entering the corresponding formally continued discrete-series string function at affine level $-k-1$. The sign restriction is preserved as well, since $A_s(j_1, m + sr) = A_s(j_1 + r/2, m)$. After applying the change of variables separately to the two formally continued discrete-series string functions, they recombine into the formally continued finite-Weyl string function at level $-k-1$. Using Eq. (18) to write the remaining Gaussian factor in terms of the fixed-charge Heisenberg character, the left branching function becomes

$$\mathcal{L}_m^{(2j_1, -k-1)}(q) = \sum_{\substack{r \in \mathbb{Z} \\ r \equiv 2j_1 - m \pmod{2}}} \eta(q) \bar{K}_{j_1 + \frac{1}{2} + \frac{kr}{2}}^{(k(k-1))}(1, q) \bar{c}_m^{(2j_1 + r, -k-1)}(q). \quad (50)$$

Here $\bar{c}_m^{(2j_1 + r, -k-1)}$ denotes the formal continuation of the finite-Weyl string function introduced above; the label $j_1 + r/2$ need not be non-negative.

For the right diagonal branching, the standard diagonal $\widehat{\mathfrak{su}}(2)$ branching gives

$$\frac{1}{\eta(q)} \Theta_{|2i-1|}^{(1)}(z_2, q) \chi_{j_2}^{(k-2)}(z_2, q) = \sum_{\substack{1 \leq n \leq k \\ n \equiv 2(i+j_2) \pmod{2}}} R_n^{(2j_2, k-1)}(q) \chi_{\frac{n}{2} - \frac{1}{2}}^{(k-1)}(z_2, q), \quad (51)$$

where

$$R_n^{(2j_2, k-1)}(q) = \sum_{a=\pm 1} a K_{(j_2 + \frac{1}{2})(k+1) - a \frac{nk}{2}}^{(k(k+1))}(1, q). \quad (52)$$

The parity condition $n \equiv 2(i+j_2) \pmod{2}$ fixes the parity of the resulting compact label for each i , while n labels the integrable $\widehat{\mathfrak{su}}(2)_{k-1}$ representation with spin $n/2 - 1/2$. When n is extended beyond its integrable range, the branching function obeys $R_{n+2(k+1)} = R_n$ and $R_{-n} = -R_n$. Hence at $n = 0$ and $n = k+1$ the branching function vanishes, while all remaining labels fold to the integrable range $n = 1, \dots, k$.

Consequently, the complete character decomposition for typical representations with finite-dimensional ground states reads

$$\chi_{[j_1, j_2]}^{(k)}(z_1, z_2, q) = \sum_{n=1}^k \sum_{\substack{m \in \mathbb{Z} \\ m+n \equiv 2(j_1+j_2) \pmod{2}}} \Psi_{\{m/2, n/2\}}^{([j_1, j_2], k)}(q) \chi_{\{m/2, n/2\}}^{(k)}(z_1, z_2, q), \quad (53)$$

where

$$\Psi_{\{m/2, n/2\}}^{([j_1, j_2], k)}(q) = (-1)^{2j_2+n+1} \mathcal{L}_m^{(2j_1, -k-1)}(q) R_n^{(2j_2, k-1)}(q). \quad (54)$$

The branching condition $m+n \equiv 2(j_1+j_2) \pmod{2}$ follows from the two intermediate parity conditions after eliminating the auxiliary label i . Likewise, $(-1)^{2i} = (-1)^{2j_2+n}$, while the additional

minus sign originates from the fermionic level-1 decomposition in Eq. (20). Using Eqs. (50) and (52), the branching function becomes

$$\begin{aligned} \Psi_{\{m/2, n/2\}}^{([j_1, j_2], k)}(q) &= (-1)^{2j_2+n+1} \sum_{\substack{r \in \mathbb{Z} \\ r \equiv 2j_1 - m \pmod{2}}} \eta(q) \bar{K}_{j_1 + \frac{1}{2} + \frac{kr}{2}}^{(k(k-1))}(1, q) \bar{c}_m^{(2j_1+r, -k-1)}(q) \\ &\quad \times \sum_{a=\pm 1} a K_{(j_2 + \frac{1}{2})(k+1) - a \frac{nk}{2}}^{(k(k+1))}(1, q). \end{aligned}$$

The decomposition obtained above is the affine counterpart of the non-affine branching rule in Eq. (13). At the level of the horizontal subalgebras, the corresponding restriction $\mathfrak{psu}(2|2) \rightarrow \mathfrak{su}(1|2)$ reduces the first bosonic $\mathfrak{su}(2)$ factor to its Cartan $\mathfrak{u}(1)$ direction, while the remaining compact $\mathfrak{su}(2)$ factor is retained in $\mathfrak{su}(1|2)$. The affine decomposition refines this finite-dimensional branching by resolving not only which $\widehat{\mathfrak{su}}(1|2)_k$ representations occur, but also the affine grades at which they occur. The factorization into L and R reflects the two bosonic parts of this branching. The function L implements the non-compact diagonal branching into $\hat{\mathfrak{u}}(1)_{-k-1}$, whereas R implements the compact diagonal branching into $\widehat{\mathfrak{su}}(2)_{k-1}$. Since both are q -dependent branching functions, their product $\Psi \propto L R$ encodes graded multiplicities over the affine current-algebra excitations rather than only the finite multiplicities of the horizontal branching.

The leading powers of the branching functions can be expressed in terms of the modular anomalies of the numerator and denominator characters. For the numerator and denominator representations one finds

$$m_{[j_1, j_2]}^{(k)} = \frac{-(2j_1+1)^2 + (2j_2+1)^2}{4k} + \frac{1}{12} \quad \text{and} \quad m_{\{m/2, n/2\}}^{(k)} = \frac{-m^2 + n^2}{4(k+1)},$$

respectively. Consequently, the lowest power of q occurring in the branching function is

$$\text{ord}_q(\Psi_{\{m/2, n/2\}}^{([j_1, j_2], k)}) = m_{[j_1, j_2]}^{(k)} - m_{\{m/2, n/2\}}^{(k)} + r_{m, n}^{j_1, j_2},$$

where

$$r_{m, n}^{j_1, j_2} = \frac{(2j_2+1-n)^2}{4} + \frac{\delta}{4} - \frac{1}{4} + \max\left(0, \frac{|m| - 2j_1 - \delta}{2}\right),$$

with $\delta \in \{0, 1\}$ determined by $\delta \equiv 2j_1 - m \equiv 2j_2 - n \pmod{2}$. The quantity $r_{m, n}^{j_1, j_2}$ is a non-negative integer and gives the affine grade at which the corresponding denominator representation first appears. The quadratic term $(2j_2+1-n)^2/4$ originates from the displacement of the compact charge lattices in the right diagonal branching (51).

As a consistency check, the terms with $r_{m, n}^{j_1, j_2} = 0$ reproduce exactly the non-affine branching rule Eq. (13). For $\delta = 0$, the condition $r_{m, n}^{j_1, j_2} = 0$ requires

$$n = 2j_2 \quad \text{or} \quad n = 2j_2 + 2, \quad |m| \leq 2j_1, \quad m \equiv 2j_1 \pmod{2}.$$

Thus $m/2 = b$, with $b = -j_1, \dots, j_1$, and these two cases give the families $\{b, j_2\}$ and $\{b, j_2+1\}$, respectively. For $\delta = 1$, one instead finds

$$n = 2j_2 + 1, \quad |m| \leq 2j_1 + 1, \quad m \equiv 2j_1 + 1 \pmod{2},$$

which gives the union of the two families $\{b + \frac{1}{2}, j_2 + \frac{1}{2}\}$ and $\{b - \frac{1}{2}, j_2 + \frac{1}{2}\}$.

The overall factor $(-1)^{2j_2+n+1}$ in Eq. (54) assigns a negative sign to the two $\delta = 0$ families and a positive sign to the $\delta = 1$ family, reproducing the parity pattern of Eq. (13). Moreover, the leading coefficients reproduce the non-affine multiplicities: for $\delta = 1$ the two positive families overlap, so that the endpoint values of m occur with coefficient one and the interior values with coefficient two. Hence the $r = 0$ part of the affine decomposition reproduces the complete horizontal branching, including its signed multiplicities. For $j_2 = 0$, the formal family with $n = 0$ is absent from the affine sum in Eq. (53), in agreement with the omission of the formal $\{b, 0\}$ term in Eq. (13). In fact, the corresponding compact branching function vanishes identically.

At the affine atypicality loci $\frac{m}{2} = \pm \frac{n}{2} + t(k+1)$, $t \in \mathbb{Z}$, the corresponding denominator Kac modules are reducible rather than irreducible typical modules. Such labels can occur in the decomposition only when $j_1 + j_2 \in \mathbb{Z}$, as follows from the branching parity $m + n \equiv 2(j_1 + j_2) \pmod{2}$. At these values, however, the character decomposition (53) does not determine the possible indecomposable module structure.

9. Decomposition of typical characters with infinite-dimensional ground states

The factorization into left and right diagonal branchings extends directly to typical numerator representations with infinite-dimensional ground states. The compact branching function $R_n^{(2j_2, k-1)}$ and the fermionic recombination are unchanged. Only the left branching is modified by replacing the finite-Weyl negative-level character with the appropriate infinite-dimensional non-compact character.

We first consider the typical discrete series representations $[(s, j_1), j_2]$, with $s = \pm 1$, $-\frac{k+1}{2} < j_1 < -\frac{1}{2}$, $j_2 = 0, \frac{1}{2}, \dots, \frac{k-2}{2}$, and $j_1 + j_2 + 1 \neq 0$. Their characters factorize according to Eq. (29), so that only the non-compact left branching differs from the finite-ground-state case. Using the discrete-series string functions derived in Sec. 7, we define the single-branch left diagonal branching function by

$$\mathcal{L}_m^{((s, j_1), -k-1)}(q) = \sum_{p \in -2sj_1 + 2\mathbb{Z}} \bar{c}_p^{((s, j_1), -k-2)}(q) \bar{\Theta}_{p(k+1)-m(k+2)}^{((k+1)(k+2))}(1, q). \quad (55)$$

This is the single-branch analogue of Eq. (47). The corresponding diagonal decomposition reads

$$\frac{1}{\eta(q)} \Theta_{2i}^{(1)}(z_1, q) \chi_{(s, j_1)}^{(-k-2)}(z_1, q) = \sum_{m \in -2sj_1 + 2i + 2\mathbb{Z}} \mathcal{L}_m^{((s, j_1), -k-1)}(q) \bar{K}_{\frac{m}{2}}^{(-k-1)}(z_1, q).$$

Unlike the finite-Weyl case, the fixed branch has the shifted affine charge lattice $-2sj_1 + 2i + 2\mathbb{Z}$, and since j_1 need not be half-integral, the denominator $u(1)$ charge $m/2$ need not be half-integral.

The compact diagonal branching is exactly Eq. (51), with the same branching function $R_n^{(2j_2, k-1)}$. Combining the two diagonal products with the unchanged fermionic recombination gives

$$\chi_{[(s, j_1), j_2]}^{(k)}(z_1, z_2, q) = \sum_{n=1}^k \sum_{m \in -2sj_1 + n - 2j_2 + 2\mathbb{Z}} \Psi_{\{\frac{m}{2}, \frac{n}{2}\}}^{([(s, j_1), j_2], k)}(q) \chi_{\{\frac{m}{2}, \frac{n}{2}\}}^{(k)}(z_1, z_2, q), \quad (56)$$

where

$$\Psi_{\{\frac{m}{2}, \frac{n}{2}\}}^{([(s, j_1), j_2], k)}(q) = (-1)^{2j_2+n+1} \mathcal{L}_m^{((s, j_1), -k-1)}(q) \mathcal{R}_n^{(2j_2, k-1)}(q).$$

We next consider the principal continuous representations $j_1 \in -\frac{1}{2} + i\mathbb{R}$, $0 \leq \alpha < 1$, and $j_2 = 0, \frac{1}{2}, \dots, \frac{k-2}{2}$. Their characters are typical and factorize according to Eq. (34). The bilateral weight distribution in Sec. 6 implies that the fixed non-compact charges of the continuous character lie in $p \in 2\alpha + 2\mathbb{Z}$. Using the principal-continuous string functions derived in Sec. 7, the left branching function $\mathcal{L}_m^{((j_1, \alpha), -k-1)}$ is obtained from the same convolution as in the discrete-series case, with $\bar{\mathcal{C}}_p^{((j_1, \alpha), -k-2)}$ and $p \in 2\alpha + 2\mathbb{Z}$. The corresponding diagonal decomposition reads

$$\frac{1}{\eta(q)} \Theta_{2i}^{(1)}(z_1, q) \chi_{(j_1, \alpha)}^{(-k-2)}(z_1, q) = \sum_{m \in 2\alpha+2i+2\mathbb{Z}} \mathcal{L}_m^{((j_1, \alpha), -k-1)}(q) \bar{\mathcal{K}}_{\frac{m}{2}}^{(-k-1)}(z_1, q).$$

Combining this left diagonal decomposition with the compact branching function $\mathcal{R}_n^{(2j_2, k-1)}$ and the same fermionic recombination gives

$$\chi_{[(j_1, \alpha), j_2]}^{(k)}(z_1, z_2, q) = \sum_{n=1}^k \sum_{m \in 2\alpha+n-2j_2+2\mathbb{Z}} \Psi_{\{\frac{m}{2}, \frac{n}{2}\}}^{([(j_1, \alpha), j_2], k)}(q) \chi_{\{\frac{m}{2}, \frac{n}{2}\}}^{(k)}(z_1, z_2, q),$$

where

$$\Psi_{\{\frac{m}{2}, \frac{n}{2}\}}^{([(j_1, \alpha), j_2], k)}(q) = (-1)^{2j_2+n+1} \mathcal{L}_m^{((j_1, \alpha), -k-1)}(q) \mathcal{R}_n^{(2j_2, k-1)}(q).$$

In both typical decompositions, the denominator character is irreducible and typical away from the affine atypicality loci $m = \pm n + 2r(k+1)$, $r \in \mathbb{Z}$. On these loci the character appearing in the decomposition is the corresponding reducible Kac character rather than the character of an irreducible typical module. The discrete-series charge lattice intersects these loci only if $j_1 + j_2 \in \mathbb{Z}$, whereas the charge lattice of a principal continuous numerator intersects them precisely when $\alpha - j_2 \in \mathbb{Z}$. Since $0 \leq \alpha < 1$, the latter condition gives $\alpha = 0$ for integral j_2 and $\alpha = \frac{1}{2}$ for half-integral j_2 . This does not contradict the fact that all principal continuous numerator representations are typical: the reducibility here concerns the corresponding $\widehat{\mathfrak{su}}(1|2)_k$ denominator modules. The character identities do not determine their possible indecomposable extension structures.

10. Decomposition of finite-ground-state atypical characters from discrete-series branches

The finite-ground-state atypical characters are reconstructed from the two discrete-series branches. For $j = \frac{1}{2}, 1, \dots, \frac{k-2}{2}$, this reconstruction is given by Eq. (26). We first decompose one fixed module $[j]_s$. After the two diagonal branchings, we isolate those terms whose denominator charges satisfy the affine atypicality condition and identify them with the irreducible atypical denominator characters of Eq. (22). The complementary terms give irreducible typical denominator characters. The two branches are combined only at the end. The exceptional vacuum case is treated separately according to Eq. (27).

For a fixed branch $s = \pm 1$, isolating the corresponding non-compact Weyl branch in Eq. (25) gives

$$\chi_{[j]_s}^{(k)}(z_1, z_2, q) = F^2(z_1, z_2, q) \frac{-i}{\theta_1(z_2, q)} \sum_{a=\pm 1} a \sum_{t \in \mathbb{Z}} \frac{\chi_{(-s, j+\frac{1}{2})}^{(-k-2)}(z_1, q) \bar{\Theta}_{2a(j+1+tk)}^{(k)}(z_2, q)}{(1 - z_1^{\frac{s}{2}} z_2^{\frac{a}{2}} q^t)^2}.$$

The non-compact factor may be written as

$$\chi_{(-s, j+\frac{1}{2})}^{(-k-2)}(z_1, q) = -\frac{is}{\theta_1(z_1, q)} \bar{\Theta}_{2s(j+1)}^{(-k)}(z_1, q).$$

This is the formal continuation of the discrete-series branch in Eq. (30) to $j_1 = j + \frac{1}{2}$. This continuation lies outside the physical unflowed discrete-series range and is used here only at the level of characters.

We use the single-branch function $L_m^{((s, j_1), -k-1)}$ defined in Eq. (55), formally continued here to $(s, j_1) = (-s, j + \frac{1}{2})$. Using the fermionic decomposition and performing the two diagonal theta products, we obtain

$$\begin{aligned} \chi_{[j]_s}^{(k)}(z_1, z_2, q) &= F(z_1, z_2, q) \frac{-i}{\eta(q)\theta_1(z_2, q)} \sum_{\substack{m, n' \in \mathbb{Z} \\ m \equiv n' \pmod{2}}} (-1)^{2j+n'} L_{sm}^{((-s, j+\frac{1}{2}), -k-1)}(q) \\ &\times \bar{K}_{(j+1)(k+1)-\frac{n'k}{2}}^{(k(k+1))}(1, q) \sum_{t \in \mathbb{Z}} \sum_{a=\pm 1} a \frac{\bar{\Theta}_{sm}^{(-k-1)}(z_1, q) \bar{\Theta}_{a(n'+2t(k+1))}^{(k+1)}(z_2, q)}{(1 - z_1^{\frac{s}{2}} z_2^{\frac{a}{2}} q^t)^2}. \end{aligned} \quad (57)$$

Here we have relabelled the left charge by $m \mapsto sm$. We keep the compact charge $n' \in \mathbb{Z}$ unfolded at this stage; affine Weyl folding is performed only after the atypical terms have been separated. Below, $n \in \mathbb{N}$ will instead denote the positive label of the resulting non-vacuum irreducible atypical denominator character; the vacuum case $n = 0$ is treated separately.

We shall also use the formal continuation of the denominator atypical characters to negative labels. From Eq. (22), using $a \mapsto -a$, $t \mapsto -t$, and $1/(1 - X^{-1}) = -X/(1 - X)$, one obtains

$$\chi_{\{-j\}_s}^{(k)} = \chi_{\{j-\frac{1}{2}\}_{-s}}^{(k)}. \quad (58)$$

This is an identity of meromorphic characters: it relates two continuations of the same meromorphic expression and should not be interpreted as an ordinary finite Weyl reflection. To extract the irreducible atypical denominator characters from Eq. (57), we expand the double pole in non-negative powers of $X = z_1^{s/2} z_2^{a/2} q^t$,

$$\frac{1}{(1 - X)^2} = \sum_{u \in \mathbb{N}_0} \frac{X^u}{1 - X}.$$

Each factor X^u shifts the two fixed-charge labels in the numerator from sm and $a(n' + 2t(k+1))$ to $s(m+u)$ and $a(n' + u + 2t(k+1))$, respectively. The atypical charge pattern requires the same shifted base charge in the two fixed-charge theta factors, and therefore fixes $n' = m$. The original

sum over m and n' thus reduces to a single sum over m . On these loci, the factor q^{tu} contained in X^u precisely accounts for the corresponding change of the Gaussian factors in the two fixed-charge theta functions. Let $n \in \mathbb{N}$ and $b = \pm 1$ denote the positive label and branch of the target denominator character $\chi_{\{n/2\}b}^{(k)}$. For fixed n and b , the pole-expansion index u is then fixed by charge matching rather than summed independently. For $b = s$, comparison with Eq. (22) gives $m + u = n + 1$. For $b = -s$, applying Eq. (58) with $j = (n + 1)/2$ gives instead $m + u = -n$. Hence $u = -m + bs n + \max(0, bs)$. It is convenient to set $p = -m$, so that the remaining sum is written with a lower bound. Since $u \geq 0$, one obtains $p \geq -bs n - \max(0, bs)$. The atypical contribution to a fixed module $[j]_s$ can therefore be written as

$$\chi_{[j]_s}^{(k)} \Big|_{\text{atyp}} = \Psi_{\{0\}}^{([j]_s, k)}(q) \chi_{\{0\}}^{(k)} + \sum_{n \in \mathbb{N}} \sum_{b=\pm 1} \Psi_{\{\frac{n}{2}\}b}^{([j]_s, k)}(q) \chi_{\{\frac{n}{2}\}b}^{(k)}. \quad (59)$$

For $n \geq 1$, $b = \pm 1$ labels two inequivalent irreducible affine atypical denominator modules, with

$$\Psi_{\{\frac{n}{2}\}b}^{([j]_s, k)}(q) = (-1)^{2j+1} \sum_{\substack{p \in \mathbb{Z} \\ p \geq -bsn - \max(0, bs)}} (-1)^p \mathbb{L}_{-sp}^{((-s, j+\frac{1}{2}), -k-1)}(q) \bar{K}_{(j+1)(k+1)+\frac{p}{2}}^{(k(k+1))}(1, q).$$

At $n = 0$, the two formal contributions instead represent the unique denominator vacuum character and must be collected into the single branching function

$$\Psi_{\{0\}}^{([j]_s, k)}(q) = (-1)^{2j+1} \sum_{b=\pm 1} \sum_{\substack{p \in \mathbb{Z} \\ p \geq -\max(0, bs)}} (-1)^p \mathbb{L}_{-sp}^{((-s, j+\frac{1}{2}), -k-1)}(q) \bar{K}_{(j+1)(k+1)+\frac{p}{2}}^{(k(k+1))}(1, q).$$

Equivalently, the vacuum coefficient may be viewed as the sum of the two formal $n = 0$ branch contributions; we display the vacuum term separately because both contributions correspond to the same irreducible denominator module, cf. Eq. (9). For fixed $n \geq 1$, the lower bound on p depends only on the relative branch sign bs . For $b = s$ it reduces to $p \geq -n - 1$, whereas for $b = -s$ it becomes $p \geq n$. These two ranges are related by the meromorphic continuation identity in Eq. (58). The factor $(-1)^p$ does not originate from the expansion of the Appell denominator. It arises solely from the relabelling $p = -m$ of the original parity factor, since $(-1)^{2j+m} = (-1)^{2j}(-1)^p$.

For the complementary typical contribution, we now return to $n = 1, \dots, k$ as the folded integrable compact label. The denominator charges on the affine atypicality loci $m = \pm n + 2v(k + 1)$, $v \in \mathbb{Z}$, have already been assigned to irreducible atypical denominator characters. The typical contribution therefore runs over the complementary charges:

$$\chi_{[j]_s}^{(k)}(z_1, z_2, q) \Big|_{\text{typ}} = \sum_{n=1}^k \sum_{\substack{m \in \mathbb{Z} \\ m \equiv n \pmod{2} \\ m \neq \pm n + 2v(k+1) \text{ for all } v \in \mathbb{Z}}} \Psi_{\{\frac{n}{2}, \frac{n}{2}\}}^{([j]_s, k)}(q) \chi_{\{\frac{n}{2}, \frac{n}{2}\}}^{(k)}(z_1, z_2, q). \quad (60)$$

To obtain these branching functions, we expand the double pole directly according to the geometric-series identity

$$\frac{1}{(1-X)^2} = \sum_{u \in \mathbb{N}_0} (u+1) X^u.$$

Recall that $n' \in \mathbb{Z}$ is the unfolded compact charge appearing in the intermediate expression, whereas $n = 1, \dots, k$ is the folded integrable compact label of the denominator representation. For each fixed u , we fold the shifted unfolded charge $n' + u$ back to the integrable range by writing

$$n' + u = d n + 2r(k+1), \quad d = \pm 1, \quad r \in \mathbb{Z},$$

or equivalently $n' = d n - u + 2r(k+1)$. Thus u remains an independent summation index, while d distinguishes the two Weyl-related orientations and r specifies the affine translate of the shifted charge $n' + u$. The compact fixed-charge label then becomes $a[dn + 2(r+t)(k+1)]$. For fixed d and r , the map $t \mapsto \tilde{t} = d(r+t)$ is a bijection of $\mathbb{Z}$, and together with $\tilde{a} = ad$ it rewrites this label as $\tilde{a}[n + 2\tilde{t}(k+1)]$. Since $a = d\tilde{a}$, the sums over a and t reconstruct the affine Weyl numerator of the compact denominator character. The resulting factor d is retained in the branching coefficient, while r remains an independent summation index. The branching function is

$$\begin{aligned} \Psi_{\{\frac{m}{2}, \frac{n}{2}\}}^{([j]_s, k)}(q) &= (-1)^{2j+n} \sum_{u \in \mathbb{N}_0} (-1)^u (u+1) \mathbb{L}_{m-su}^{((-s, j+\frac{1}{2}), -k-1)}(q) \\ &\quad \times \sum_{d=\pm 1} d \sum_{r \in \mathbb{Z}} \bar{K}_{(j+1)(k+1) - \frac{dn-u+2r(k+1)}{2}}^{(k(k+1))}(1, q) q^{\frac{sm-dn-2r(k+1)}{2(k+1)}u}. \end{aligned}$$

Unlike in the atypical contribution above, the double-pole expansion index $u \in \mathbb{N}_0$ remains to be summed over and is not fixed by charge matching. The factor $q^{(sm-dn-2r(k+1))u/[2(k+1)]}$ arises because the same power X^u shifts both fixed-charge theta factors. Since their levels are $-(k+1)$ and $+(k+1)$, the terms quadratic in u cancel, leaving only the displayed term linear in u . This couples the non-compact and compact charge sectors, so that the branching function no longer factorizes into independent L and R factors.

We may now reconstruct the finite-ground-state atypical character itself. For $j = \frac{1}{2}, 1, \dots, \frac{k-2}{2}$, one has

$$\begin{aligned} \chi_{[j]}^{(k)}(z_1, z_2, q) &= \Psi_{\{0\}}^{([j], k)}(q) \chi_{\{0\}}^{(k)}(z_1, z_2, q) + \sum_{n \in \mathbb{N}} \sum_{b=\pm 1} \Psi_{\{\frac{n}{2}\}_b}^{([j], k)}(q) \chi_{\{\frac{n}{2}\}_b}^{(k)}(z_1, z_2, q) \\ &\quad + \sum_{n=1}^k \sum_{\substack{m \in \mathbb{Z} \\ m \equiv n \pmod{2} \\ m \neq \pm n + 2v(k+1) \text{ for all } v \in \mathbb{Z}}} \Psi_{\{\frac{m}{2}, \frac{n}{2}\}}^{([j], k)}(q) \chi_{\{\frac{m}{2}, \frac{n}{2}\}}^{(k)}(z_1, z_2, q). \end{aligned}$$

Since the two modules $[j]_s$ have been expressed in the same denominator character basis, the branching functions are reconstructed linearly. For every denominator label λ occurring above,

$$\Psi_{\lambda}^{([j], k)}(q) = - \sum_{s=\pm 1} \Psi_{\lambda}^{([j]_s, k)}(q).$$

By Eq. (26), this reconstructs the finite-ground-state atypical character (25).

The vacuum is exceptional. According to Eq. (27), either formal branch continuation $s = \pm 1$ may be used, but the two branches are not added. Hence, for every denominator label λ in the fixed-branch decomposition,

$$\Psi_{\lambda}^{([0], k)}(q) = - \Psi_{\lambda}^{([j]_s, k)}(q) \Big|_{j=-\frac{1}{2}}.$$

In particular, every factor $\mathbb{L}_{m-su}^{((-s, j+\frac{1}{2}), -k-1)}$ is replaced by $\mathbb{L}_{m-su}^{((-s, 0), -k-1)}$. The same replacement applies to the atypical term with lower label $-sp$. Thus the fixed-branch formulas above continue directly to the vacuum, with the chosen branch expansion understood throughout.

As an independent low-order consistency check, the direct expansion of the affine vacuum character in Sec. 5, Eq. (28), gives $-q^{1/12}\{0\}$ at the ground-state level and $q^{13/12}(-\{\frac{1}{2}\}_+ - \{\frac{1}{2}\}_- + \{0, 1\})$ at the first excited affine grade. The continuation above reproduces precisely these denominator characters and coefficients.

The geometric interpretation is the same as for the branching discussed in Secs. 2 and 8. The denominator algebra $\mathfrak{su}(1|2)$ is the stabilizer algebra of the coset geometry, and the branching describes the corresponding restriction from $\mathfrak{psu}(1,1|2)$ to $\mathfrak{su}(1|2)$. The affine branching functions encode the graded multiplicities associated with the denominator representations. In the atypical case considered here, the decomposition also separates the generic typical denominator representations from the irreducible atypical ones at the affine atypicality loci.

11. Spectral flow

We consider the spectral-flow automorphisms $\gamma_{\pm} = \gamma^{(\mp 1, 1)}$ of Ref. [GQS07, arXiv, Eqs. (3.11)–(3.13) and the discussion following Eq. (3.33)]. More generally, the affine algebra admits spectral flows $\gamma^{(w_1, w_2)}$. Their action shifts the fermionic mode numbers by $(w_1 + w_2)/2$ or $(w_1 - w_2)/2$, which are integers for $\gamma_{\pm}$. Thus, these flows preserve the integer fermionic moding and do not interpolate between the Ramond and Neveu-Schwarz sectors. In the following we mainly use $\gamma_-^w = \gamma^{(w, w)}$. Its w -th power acts on characters as

$$\gamma_-^w[\chi](z_1, z_2, q) = z_1^{-\frac{wk}{2}} z_2^{\frac{wk}{2}} \chi(q^w z_1, q^w z_2, q), \quad w \in \mathbb{Z}.$$

For a discrete series subcharacter, with $s = \pm 1$ labelling the two branches, one finds

$$z^{-\frac{wk}{2}} \chi_{(s, j_1)}^{(-k-2)}(q^w z, q) = (-1)^w z^w q^{\frac{w^2(k+2)}{4}} \chi_{(s, j_1 + s\frac{wk}{2})}^{(-k-2)}(z, q).$$

Similarly, the compact factor obeys

$$z^{\frac{wk}{2}} \chi_{j_2}^{(k-2)}(q^w z, q) = (-1)^w z^w q^{-\frac{w^2(k-2)}{4}} \chi_{j_2 + \frac{wk}{2}}^{(k-2)}(z, q), \quad (61)$$

where the shifted compact label is understood through its affine Weyl continuation. Finally,

$$F^2(q^w z_1, q^w z_2; q) = z_1^{-w} z_2^{-w} q^{-w^2} F^2(z_1, z_2; q).$$

The automorphy factors cancel since $\frac{w^2(k+2)}{4} - \frac{w^2(k-2)}{4} - w^2 = 0$. Since the spectral-flow number will be used below as an explicit sector label, we define

$$\chi_{[(s, j_1), j_2]_w}^{(k)} := \gamma_-^w[\chi_{[(s, j_1), j_2]}^{(k)}] = \chi_{[(s, j_1 + s\frac{wk}{2}), j_2 + \frac{wk}{2}]}^{(k)},$$

where the labels in the last expression are unfolded labels and need not lie in the physical unflowed range. Thus, for the discrete series the spectral-flow label w can equivalently be represented by

shifts of the bosonic labels, but we retain it explicitly in order to distinguish the spectral-flow sectors appearing in the modular transformations and in the state space.

For one unit of spectral flow, folding the bosonic labels back to the standard ranges gives for the typical discrete series

$$[(\pm, j_1), j_2] \xrightarrow{\gamma_{\mp}} [(\mp, -j_1 - \frac{k}{2} - 1), -j_2 + \frac{k}{2} - 1].$$

The unfolded and folded forms are equivalent at the character level: the non-compact and compact affine Weyl reflections each contribute a minus sign, which cancel in the full supercharacter. For the atypical discrete series representations the corresponding relation is

$$[j]_{\pm} \xrightarrow{\gamma_{\mp}} [-j + \frac{k}{2} - 1]_{\mp}.$$

For general winding number it is preferable to retain the spectral flow explicitly rather than to identify every flowed atypical module with a formally continued unflowed label.

For the principal continuous series, spectral flow cannot in general be written as a simple shift of the unflowed representation labels, unlike in the discrete case. This can already be seen from the bilateral weight distribution in Eq. (6). Under $z \rightarrow q^w z$, each term $z^{\alpha+n}$ acquires an additional factor $q^{w(\alpha+n)}$. Thus the different weights of the representation acquire different powers of q . A shift of α only shifts the set of weights and cannot remove this n -dependent factor. A shift of j_1 cannot do so either: according to Eq. (44), j_1 enters through the factor $q^{-(j_1+1/2)^2/k}$, which is the same for all weights. Hence the flowed character cannot be identified with an unflowed principal continuous character with shifted j_1 and α . We therefore retain the winding number explicitly and define

$$\chi_{[(j_1, \alpha), j_2]_w}^{(k)}(z_1, z_2, q) := \gamma_w^{-1} [\chi_{[(j_1, \alpha), j_2]}^{(k)}](z_1, z_2, q) = z_1^{-\frac{wk}{2}} z_2^{\frac{wk}{2}} \chi_{[(j_1, \alpha), j_2]}^{(k)}(q^w z_1, q^w z_2, q).$$

Thus, we retain an explicit spectral-flow label w for both the discrete and principal continuous series in what follows. In both cases the complete spectral-flow orbit is required for modular closure, as discussed in Sec. 13. For the discrete series the label w may equivalently be represented by the unfolded shifts of the bosonic spins given above, whereas no analogous identification with shifted unflowed labels is available for the principal continuous series.

12. Modular transformations of typical characters

The modular transformation properties determine how the discrete and principal continuous characters, together with their spectral-flow images, transform under the modular group. They are therefore needed to establish modular closure of the typical characters considered below and provide ingredients for the eventual construction of a modular-invariant torus partition function. In the sR sector considered below, the T -transformation acts diagonally on the characters, whereas the non-trivial mixing relevant for modular closure occurs under S . We therefore focus on the S -transformation below.

For modular transformations we switch to additive variables,

$$z_1 = e^{2\pi i x}, \quad z_2 = e^{2\pi i y}, \quad q = e^{2\pi i \tau}.$$

For a single theta function we use $z = e^{2\pi i x}$. In these variables the level- k theta function defined in Eq. (17) becomes

$$\Theta_{2j}^{(k)}(x, \tau) = \sum_{t \in \mathbb{Z}} e^{2\pi i k(t + \frac{j}{k})x} e^{2\pi i k(t + \frac{j}{k})^2 \tau}.$$

Its modular S -transform is

$$\Theta_{2j}^{(k)}\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) = \frac{\sqrt{-i\tau}}{\sqrt{2k}} e^{\frac{\pi i k x^2}{2\tau}} \sum_{m=0}^{2k-1} e^{-\frac{2\pi i j m}{k}} \Theta_m^{(k)}(x, \tau).$$

The integrable $\widehat{\mathfrak{su}}(2)_k$ characters transform as

$$\chi_j^{(k)}\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) = \sqrt{\frac{2}{k+2}} e^{\frac{\pi i k x^2}{2\tau}} \sum_{2j'=0}^k \sin\left(\frac{\pi(2j+1)(2j'+1)}{k+2}\right) \chi_{j'}^{(k)}(x, \tau).$$

The Jacobi theta functions transform under S as [DLM26, Sec. 20.7(viii)]

$$\theta_1\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) = -i\sqrt{-i\tau} e^{\frac{\pi i x^2}{\tau}} \theta_1(x, \tau), \quad (62)$$

$$(\theta_2, \theta_3, \theta_4)\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) = \sqrt{-i\tau} e^{\frac{\pi i x^2}{\tau}} (\theta_4, \theta_3, \theta_2)(x, \tau). \quad (63)$$

The Dedekind eta function transforms as [DLM26, Eq. (23.18.5)]

$$\eta\left(-\frac{1}{\tau}\right) = \sqrt{-i\tau} \eta(\tau).$$

For the negative-level fixed-charge Gaussian characters the real-axis Gaussian integrals are not convergent for $\text{Im } \tau > 0$. We therefore define them by analytic continuation from the convergent Gaussian integral, using the principal square-root branch, so that $\sqrt{-i\tau} > 0$ for $\tau \in i\mathbb{R}^+$. We denote this prescription by $\int_{\mathbb{R}}^{\text{F}}$ and normalize it by

$$\int_{\mathbb{R}}^{\text{F}} du e^{-\frac{2\pi i \tau}{k} u^2} = -i\sqrt{\frac{k}{-2i\tau}}. \quad (64)$$

Applying this prescription to the fixed-charge function $\bar{K}_j^{(-k)}$ defined in Eq. (18) gives

$$\bar{K}_j^{(-k)}\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) = i\sqrt{\frac{2}{k}} e^{-\frac{\pi i k x^2}{2\tau}} \int_{\mathbb{R}}^{\text{F}} dj' e^{\frac{4\pi i j j'}{k}} \bar{K}_{j'}^{(-k)}(x, \tau). \quad (65)$$

In the following, the non-compact discrete characters are transformed in the analytically continued meromorphic sense associated with the Fresnel prescription defined in Eq. (64). For comparison, in the distributional treatment of Lorentzian $\widehat{\mathfrak{sl}}(2, \mathbb{R})$ characters the generalized S -matrix contains an off-diagonal block mixing discrete and continuous representations [BN11]. The present analysis instead uses the analytically continued meromorphic/Fresnel prescription.

The modular S -transformation permutes the spin structures. Using the standard correspondence $sR \sim \theta_1$, $R \sim \theta_2$, $NS \sim \theta_3$, and $sNS \sim \theta_4$, Eqs. (62) and (63) give

$$sR \longrightarrow sR, \quad R \longleftrightarrow sNS, \quad NS \longrightarrow NS.$$

Throughout this work χ denotes the sR supercharacter, so the modular transformations below are written in the S -invariant sR sector.

Using the Fresnel prescription defined in Eq. (64), the discrete-series characters transform as

$$\chi_{(s,j)}^{(-k-2)}\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) = -\sqrt{\frac{2}{k}} e^{-\frac{\pi i(k+2)x^2}{2\tau}} \int_{\mathbb{R}}^F dj' e^{\frac{\pi i}{k}(2j+1)(2j'+1)} \chi_{(s,j')}^{(-k-2)}(x, \tau), \quad s = \pm 1. \quad (66)$$

The modular transform therefore preserves the discrete-series branch s at this unfolded level. The unfolded Fourier integral may be decomposed into the physical discrete-series interval and its spectral-flow translates, $j' \mapsto j' + s\frac{wk}{2}$, $-\frac{k+1}{2} < j' < -\frac{1}{2}$, $w \in \mathbb{Z}$. One thus obtains

$$\begin{aligned} \chi_{(s,j)}^{(-k-2)}\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) &= -\sqrt{\frac{2}{k}} e^{-\frac{\pi i(k+2)x^2}{2\tau}} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj' \sum_{w \in \mathbb{Z}} \\ &\times e^{\frac{\pi i}{k}(2j+1)(2j'+1)} e^{\pi i s(2j+1)w} \chi_{(s,j'+s\frac{wk}{2})}^{(-k-2)}(x, \tau). \end{aligned}$$

Here the sum over w decomposes the original $j' \in \mathbb{R}$ integration range into spectral-flow sectors. The integral over the finite physical interval is an ordinary integral; the resulting w -sum is understood as the decomposition of the Fresnel-prescribed unfolded integral in Eq. (66).

For later combination with the non-compact factor, the compact S -transform may equivalently be written, for any fixed $w \in \mathbb{Z}$, as

$$\chi_j^{(k-2)}\left(\frac{y}{\tau}, -\frac{1}{\tau}\right) = \sqrt{\frac{2}{k}} e^{\frac{\pi i(k-2)y^2}{2\tau}} \sum_{2j'=0}^{k-2} \sin\left(\frac{\pi}{k}(2j+1)(2j'+1)\right) (-1)^{(2j+1)w} \chi_{j'+\frac{wk}{2}}^{(k-2)}(y, \tau).$$

This is not an additional spectral-flow sum. Rather, for each fixed w it follows from the usual periodicity and Weyl-reflection identities $\chi_{j+k}^{(k-2)} = \chi_j^{(k-2)}$ and $\chi_{j+k/2}^{(k-2)} = -\chi_{k/2-1-j}^{(k-2)}$, together with a reindexing of the complete integrable j' -sum. Combining the non-compact and compact transformations with the fermionic factor, one obtains

$$\begin{aligned} \chi_{[(s,j_1),j_2]}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) &= -\frac{2}{k} e^{\frac{\pi i k}{2\tau}(y^2-x^2)} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \\ &\times \sin\left(\frac{\pi}{k}(2j_2+1)(2j'_2+1)\right) e^{\frac{\pi i}{k}(2j_1+1)(2j'_1+1)} e^{2\pi i(j_2+s j_1)w} \\ &\times \gamma_-^w \left[\chi_{[(s,j'_1),j'_2]}^{(k)}(x, y, \tau) \right]. \end{aligned} \quad (67)$$

For the continuous series the charge sum is understood distributionally. Poisson resummation, i.e. rewriting the charge-lattice sum as a Fourier sum over the dual integer lattice, together with

the Gaussian transform along $j' \in -\frac{1}{2} + i\mathbb{R}$, gives

$$\begin{aligned} \chi_{(j,\alpha)}^{(-k-2)}\left(\frac{x}{\tau}, -\frac{1}{\tau}\right) &= \sqrt{\frac{2}{k}} \int_{-\frac{1}{2}+i\mathbb{R}} dj' \int_0^1 d\alpha' \sum_{w \in \mathbb{Z}} \\ &\times e^{\frac{\pi i}{k}(2j+1)(2j'+1)} e^{-2\pi i \alpha w} e^{-\frac{\pi i(k+2)}{2\tau}(x+w\tau)^2} \chi_{(j',\alpha')}^{(-k-2)}(x+w\tau, \tau). \end{aligned}$$

The resummed dual-lattice index $w \in \mathbb{Z}$ becomes the spectral-flow label when the non-compact, compact and fermionic factors are recombined below. For the compact factor, the Weyl folding used above may be combined with the spectral-flow quasi-periodicity in Eq. (61). For any fixed $w \in \mathbb{Z}$ one obtains

$$\begin{aligned} \chi_j^{(k-2)}\left(\frac{y}{\tau}, -\frac{1}{\tau}\right) &= \sqrt{\frac{2}{k}} \sum_{2j'=0}^{k-2} \sin\left(\frac{\pi}{k}(2j+1)(2j'+1)\right) \\ &\times (-1)^{2jw} e^{\frac{\pi i(k-2)}{2\tau}(y+w\tau)^2} \chi_{j'}^{(k-2)}(y+w\tau, \tau). \end{aligned}$$

Combining the non-compact and compact factors with

$$F^2(x, y; \tau) = e^{2\pi i w(x+y) + 2\pi i \tau w^2} F^2(x+w\tau, y+w\tau; \tau),$$

all w -dependent variable factors assemble into the spectral-flowed full character. One obtains

$$\begin{aligned} \chi_{[(j_1, \alpha), j_2]}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) &= \frac{2}{k} e^{\frac{\pi i k}{2\tau}(y^2 - x^2)} \int_{-\frac{1}{2}+i\mathbb{R}} dj'_1 \int_0^1 d\alpha' \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \\ &\times e^{\frac{\pi i}{k}(2j_1+1)(2j'_1+1)} \sin\left(\frac{\pi}{k}(2j_2+1)(2j'_2+1)\right) \\ &\times e^{2\pi i(j_2-\alpha)w} \gamma_-^w [\chi_{[(j'_1, \alpha'), j'_2]}^{(k)}(x, y, \tau)]. \end{aligned} \quad (68)$$

Finally, the typical $\widehat{\mathfrak{su}}(1|2)_k$ characters transform according to the Fourier-sine kernel of Ref. [SS07, arXiv, Eq. (3.11)], with the Gaussian prescription adapted to the convention used here:

$$\chi_{\{b,j\}}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) = -\frac{2i}{k+1} e^{\frac{\pi i k}{2\tau}(y^2 - x^2)} \int_{\mathbb{R}} db' \sum_{2j'=1}^k e^{\frac{4\pi i b b'}{k+1}} \sin\left(\frac{4\pi j j'}{k+1}\right) \chi_{\{b', j'\}}^{(k)}(x, y, \tau). \quad (69)$$

The b' -integral is understood with the Fresnel prescription defined in Eq. (64).

13. Modular closure of the typical bulk sectors

Following Ref. [GQS07, arXiv, Sec. 6], we consider the auxiliary bulk state space $\hat{\mathcal{H}}_0$ of the decoupled free-field theory. This auxiliary space should not be identified with the full interacting bulk state space $\hat{\mathcal{H}}$. In the typical sector, the two are related by the intertwining construction described in Ref. [GQS07, arXiv, Sec. 6], whereas the atypical sector of the interacting theory

has a non-factorizing indecomposable structure. In this section we do not attempt to construct the latter. Instead, we use the modular transformations derived in Sec. 12 to test, within the analytically continued meromorphic/Fresnel prescription described there, the modular closure of the typical discrete and continuous sectors of $\hat{\mathcal{H}}_0$, with the usual Jacobi covariance for non-vanishing chemical potentials.

We use the notation of the preceding sections, with j_1 denoting the non-compact and j_2 the compact spin. For a left-moving chiral $\widehat{\mathfrak{psu}}(1,1|2)_k$ module $\mathcal{V}_\lambda$ with representation label λ , we denote its charge-conjugate right-moving module by $\mathcal{V}_\lambda^+$. The typical part of the auxiliary bulk state space is built from affine modules whose non-compact ground states belong to the unitary discrete and principal continuous series of $\mathfrak{sl}(2, \mathbb{R})$, together with the integrable $\widehat{\mathfrak{su}}(2)_{k-2}$ representations. The finite-dimensional non-compact ground-state representations considered in the preceding sections do not occur in this unitary bulk spectrum; they were retained there because they are relevant to the algebraic branching problem and to the construction of the coset characters. Accordingly,

$$\begin{aligned} \hat{\mathcal{H}}_0^{\text{typ}} \simeq & \bigoplus_{2j_2=0}^{k-2} \bigoplus_{w \in \mathbb{Z}} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj_1 \mathcal{V}_{[(+,j_1),j_2]_w} \otimes (\mathcal{V}_{[(+,j_1),j_2]_w})^+ \\ & \oplus \bigoplus_{2j_2=0}^{k-2} \bigoplus_{w \in \mathbb{Z}} \int_{-\frac{1}{2}+i\mathbb{R}}^1 dj_1 \int_0^1 d\alpha \mathcal{V}_{[(j_1,\alpha),j_2]_w} \otimes (\mathcal{V}_{[(j_1,\alpha),j_2]_w})^+. \end{aligned}$$

The integral over the discrete-series label j_1 is understood over the open interval $-\frac{k+1}{2} < j_1 < -\frac{1}{2}$. At the isolated values $j_1 + j_2 + 1 = 0$, the corresponding Kac modules become reducible and hence atypical. Since these values form a set of measure zero in the j_1 -integration, omitting them does not change the direct-integral expression used below.

For the discrete series, the spectral-flow notation is related to the spectral-flow automorphism introduced in Sec. 11 by $\chi_{[(+,j_1),j_2]_w}^{(k)} = \gamma_-^w [\chi_{[(+,j_1),j_2]}^{(k)}] = \chi_{[(+,j_1+wk/2),j_2+wk/2]}^{(k)}$, where the labels in the last expression are understood in the unfolded sense. It is sufficient to choose the + discrete series as an orbit representative, since its complete spectral-flow orbit also contains the opposite discrete branch after folding back to the physical representation domain. The corresponding diagonal sR character sum, representing the left-right paired contribution of this typical discrete sector, is

$$Z_{0,\text{sR}}^{\text{discrete}}(x, y, \tau) = \sum_{2j_2=0}^{k-2} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj_1 \sum_{w \in \mathbb{Z}} |\gamma_-^w [\chi_{[(+,j_1),j_2]}^{(k)}](x, y, \tau)|^2, \quad (70)$$

where the j_1 -integration is again understood over the open physical interval. Here and below, $|\chi|^2$ denotes the corresponding charge-conjugate left-right pairing associated with $\mathcal{V}_\lambda \otimes (\mathcal{V}_\lambda)^+$. For completeness, the transformation of an already spectral-flowed discrete character may be written explicitly. Using the spectral-flow automorphism together with the simultaneous integer quasi-periodicity

$$\chi_{[(s,j'_1),j'_2]}^{(k)}(x-w, y-w, \tau) = e^{2\pi i(j'_2+s j'_1)w} \chi_{[(s,j'_1),j'_2]}^{(k)}(x, y, \tau),$$

which is needed to shift the transformed arguments $x - w$ and $y - w$ back to x and y , one obtains the S -transform of the character in spectral-flow sector w ,

$$\begin{aligned} \gamma_-^w [\chi_{[(s, j_1), j_2]}^{(k)}] \left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau} \right) &= -\frac{2}{k} e^{\frac{\pi i k}{2\tau} (y^2 - x^2)} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w' \in \mathbb{Z}} \\ &\times e^{\frac{\pi i}{k} (2j_1+1)(2j'_1+1)} \sin \left(\frac{\pi}{k} (2j_2+1)(2j'_2+1) \right) \\ &\times e^{2\pi i [(j_2+s j_1)w' + (j'_2+s j'_1)w]} \gamma_-^{w'} [\chi_{[(s, j'_1), j'_2]}^{(k)}] (x, y, \tau). \end{aligned}$$

Within the meromorphic/Fresnel prescription used here, this formula shows explicitly that the complete discrete spectral-flow orbit closes under the S -transformation. This statement is prescription-dependent: in the distributional Lorentzian treatment, the generalized S -matrix also contains discrete-continuous mixing [BN11]; cf. also Sec. 12. For the diagonal character sum in Eq. (70), the unitarity of the transformation involves the integral over the physical j'_1 interval together with the sum over all spectral-flow sectors w' . The sum over w' reconstructs the unfolded real j'_1 integration from the physical interval, while the compact affine Weyl reindexing only permutes the integrable j'_2 labels up to signs. Together with the unitarity of the compact sine matrix, this gives a unitary transformation of the complete set of discrete labels.

In the diagonal left-right pairing, the overall minus signs from the holomorphic and antiholomorphic discrete S -kernels cancel. Hence, within this prescription, the diagonal discrete sR character sum in Eq. (70) is formally S -invariant at $x = y = 0$, where the Jacobi factor becomes unity. For non-vanishing chemical potentials, it instead transforms covariantly with the corresponding Jacobi factor. The specialization $x = y = 0$ is not an independent test of the kernel normalization.

We now turn to the principal continuous sector. The corresponding diagonal sR character sum is

$$\mathbb{Z}_{0, \text{sR}}^{\text{continuous}}(x, y, \tau) = \sum_{2j_2=0}^{k-2} \int_{-\frac{1}{2}+i\mathbb{R}} \frac{dj_1}{i} \int_0^1 d\alpha \sum_{w \in \mathbb{Z}} |\gamma_-^w [\chi_{[(j_1, \alpha), j_2]}^{(k)}] (x, y, \tau)|^2.$$

The continuous integral and the character sum are understood in the distributional sense used in Sec. 12. To verify modular closure explicitly, we first determine the S -transformation of an already spectral-flowed character. Using the spectral-flow automorphism together with the continuous transformation derived in Sec. 12, one obtains

$$\begin{aligned} \gamma_-^w [\chi_{[(j_1, \alpha), j_2]}^{(k)}] \left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau} \right) &= \frac{2}{k} e^{\frac{\pi i k}{2\tau} (y^2 - x^2)} \int_{-\frac{1}{2}+i\mathbb{R}} dj'_1 \int_0^1 d\alpha' \sum_{2j'_2=0}^{k-2} \sum_{w' \in \mathbb{Z}} \\ &\times e^{\frac{\pi i}{k} (2j_1+1)(2j'_1+1)} \sin \left(\frac{\pi}{k} (2j_2+1)(2j'_2+1) \right) \\ &\times e^{2\pi i [(j_2-\alpha)w' + (j'_2-\alpha')w]} \gamma_-^{w'} [\chi_{[(j'_1, \alpha'), j'_2]}^{(k)}] (x, y, \tau). \end{aligned}$$

After factoring out the common prefactor $e^{\frac{\pi i k}{2\tau} (y^2 - x^2)}$, the transformation in the representation labels is unitary with respect to the continuous measures and discrete sums appearing in the diagonal character sum. Writing $j_1 = -\frac{1}{2} + ip$ and $j'_1 = -\frac{1}{2} + ip'$, the non-compact factor becomes $e^{-4\pi i p p' / k}$. Splitting the overall normalization as $2/k = \sqrt{2/k} \sqrt{2/k}$, the first factor $\sqrt{2/k}$ gives

the standard unitary Fourier transform on $\mathbb{R}$. Since $dj'_1 = i dp'$, changing from the oriented contour measure to the real measure introduces only an overall phase, which drops out of the left-right pairing. The second factor $\sqrt{2/k}$ combines with the sine function to give the unitary $\widehat{\mathfrak{su}}(2)_{k-2}$ S -matrix. Finally, α , defined modulo integers, and $w \in \mathbb{Z}$ are Fourier-dual variables through the corresponding phase factors. Thus the complete set of continuous spectral-flow sectors closes under S . As in the discrete case, the corresponding diagonal sR character sum transforms covariantly at generic chemical potentials with the appropriate Jacobi factor. The specialization $x = y = 0$ is subject to the same meromorphic/distributional qualification and is not needed for the closure argument.

The T -transformation is diagonal in the representation labels and acts on each sR character only by its conformal phase. In the diagonal left-right pairing this phase cancels between the holomorphic and antiholomorphic factors, so that both character sums are T -invariant. Together with the discrete and continuous S -transformations above, this establishes the modular closure of the typical part of the auxiliary bulk state space once the full spectral-flow orbits are included. The result provides a consistency check of the modular kernels derived in the preceding section; it does not constitute a construction of the complete interacting torus partition function.

14. Modular transformation of $\widehat{\mathfrak{su}}(1|2)_k$ atypical characters

The irreducible atypical $\widehat{\mathfrak{su}}(1|2)_k$ supercharacters occur explicitly in the denominator decomposition of the finite-ground-state atypical numerator characters derived in Sec. 10. For the modular transformation of the corresponding branching functions, it is useful to express them in the same $\{b, j\}$ character basis used for the generic denominator characters. We therefore derive their modular transformation from a formal affine resolution. Higher-level Appell representations and their modular transformations provide an alternative description [STT05, SS07]. At $k = 1$ and for vanishing chemical potentials, corresponding in our notation to $x = y = 0$, Ref. [SS07] makes the relation between the real-contour and Mordell representations explicit by deforming the contour and accounting for the crossed poles. To our knowledge, an explicit general- k contour identification with the resolved Fresnel-Abel representation has not been given. The general higher-level Appell transformation is nevertheless available from Ref. [STT05], while such a contour identification is not required for the derivation based on the formal resolution used below.

The affine analogue of the finite-dimensional character relation in Eq. (12) is

$$\chi_{\{j\}_s}^{(k)} = - \sum_{n=1}^{\infty} \chi_{\{s(j+\frac{n}{2}), j+\frac{n}{2}\}}^{(k)}, \quad s = \pm 1. \quad (71)$$

When the labels on the right-hand side leave the unflowed integrable range, the character formula in Eq. (19) is understood by formal continuation. Each term lies on an affine atypicality locus and hence corresponds there to a reducible Kac module. Eq. (71) is an identity of supercharacters, not a direct-sum decomposition of modules. Applying the Kac-character transformation in Eq. (69)

term by term to Eq. (71) gives

$$\begin{aligned} \chi_{\{j\}s}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) &= \frac{2i}{k+1} e^{\frac{\pi i k}{2\tau}(y^2-x^2)} \int_{\mathbb{R}}^F db' \sum_{2j'=1}^k \sum_{n=1}^{\infty} \\ &\times e^{\frac{4\pi i}{k+1}s(j+\frac{n}{2})b'} \sin\left(\frac{4\pi}{k+1}\left(j+\frac{n}{2}\right)j'\right) \chi_{\{b',j'\}}^{(k)}(x, y, \tau). \end{aligned} \quad (72)$$

The overall sign is positive because the minus sign in the resolution cancels that in Eq. (69). Using

$$\sin\left(\frac{4\pi}{k+1}\left(j+\frac{n}{2}\right)j'\right) = \frac{1}{2i} \sum_{a=\pm 1} a e^{\frac{4\pi i}{k+1}a(j+\frac{n}{2})j'},$$

the two exponential factors in Eq. (72) combine naturally if we define $X_a := sb' + aj'$. The part of the integrand depending on the representation labels then becomes

$$\frac{1}{k+1} \sum_{a=\pm 1} a e^{\frac{4\pi i}{k+1}jX_a} \sum_{n=1}^{\infty} e^{\frac{2\pi i}{k+1}nX_a}.$$

For real b' , X_a is real, and hence the common ratio $r_a := e^{\frac{2\pi i}{k+1}X_a}$ of the geometric series satisfies $|r_a| = 1$. The series therefore does not in general converge in the ordinary sense. We introduce a damping factor $e^{-\varepsilon n}$, $\varepsilon > 0$, for which

$$\sum_{n=1}^{\infty} e^{-n\varepsilon + \frac{2\pi i}{k+1}nX_a} = \frac{e^{-\varepsilon + \frac{2\pi i}{k+1}X_a}}{1 - e^{-\varepsilon + \frac{2\pi i}{k+1}X_a}},$$

and define the original series by the Abel limit $\varepsilon \rightarrow 0^+$. Equivalently, $e^{-\varepsilon + \frac{2\pi i}{k+1}X_a} = e^{\frac{2\pi i}{k+1}(X_a + i\frac{k+1}{2\pi}\varepsilon)}$, so for $\varepsilon > 0$ the regularization amounts to approaching the real X_a axis from the upper half-plane. This prescription becomes relevant precisely at those values of X_a for which the denominator vanishes in the limit. We denote the resulting coefficient of $\chi_{\{b',j'\}}^{(k)}$ by the generalized S -kernel

$$\mathcal{S}_{\{b',j'\}}^{F,(\{j\}_s,k)} := \lim_{\varepsilon \rightarrow 0^+} \frac{1}{k+1} \sum_{a=\pm 1} a e^{\frac{4\pi i}{k+1}j(sb'+aj')} \frac{e^{-\varepsilon + \frac{2\pi i}{k+1}(sb'+aj')}}{1 - e^{-\varepsilon + \frac{2\pi i}{k+1}(sb'+aj')}}. \quad (73)$$

The superscript F refers to the Fresnel prescription for the b' -integration defined in Eq. (64), while the Abel limit specifies how the singular values of the geometric sum are approached. With this notation,

$$\chi_{\{j\}s}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) = e^{\frac{\pi i k}{2\tau}(y^2-x^2)} \int_{\mathbb{R}}^F db' \sum_{2j'=1}^k \mathcal{S}_{\{b',j'\}}^{F,(\{j\}_s,k)} \chi_{\{b',j'\}}^{(k)}(x, y, \tau). \quad (74)$$

The poles of the kernel follow directly from the denominator of the geometric sum above. In the Abel limit $\varepsilon \rightarrow 0^+$, it vanishes when $e^{\frac{2\pi i}{k+1}X_a} = 1$, that is, when $X_a = m(k+1)$ with $m \in \mathbb{Z}$. Since $X_a = sb' + aj'$, this gives $b' = s(m(k+1) - aj')$. For $s, a = \pm 1$, these values are, modulo affine translations, precisely $b' = \pm j' \bmod k+1$. These are the affine atypicality loci in

the (b', j') labels: away from them the corresponding Kac modules are irreducible and typical, whereas on them they are atypical and reducible. Thus the poles introduced by resumming the affine resolution occur exactly at the reducible Kac labels.

As a useful check, consider the vacuum $j = 0$. Since the two branches $\{0\}_\pm$ coincide, the resulting kernel is independent of s . Using

$$\frac{e^{2iu}}{1 - e^{2iu}} = -\frac{1}{2} + \frac{i}{2} \cot u,$$

the constant terms cancel in the sum over a . Away from its poles, and with the Abel boundary value defined in Eq. (73) understood on the real contour, the kernel can therefore be written as

$$\mathcal{S}_{\{b', j'\}}^{\text{F}, (\{0\}, k)} = -\frac{i}{2(k+1)} \frac{\sin\left(\frac{2\pi j'}{k+1}\right)}{\sin\left(\frac{\pi(b'+j')}{k+1}\right) \sin\left(\frac{\pi(b'-j')}{k+1}\right)}. \quad (75)$$

For $k = 1$ the spin sum contains only $j' = \frac{1}{2}$, so that

$$\mathcal{S}_{\{b', \frac{1}{2}\}}^{\text{F}, (\{0\}, 1)} = \frac{i}{2 \cos(\pi b')}. \quad (76)$$

The characteristic $1/\cos(\pi b')$ dependence and its pole locations agree with the level-one vacuum transformation of Ref. [SS07, arXiv, Eq. (3.12)]. The opposite overall sign originates from the different prescription for the formally divergent Gaussian integral: Ref. [SS07, arXiv, Eq. (3.11)] uses the opposite branch of the Gaussian continuation compared with the Fresnel prescription adopted here.

Eq. (74) should therefore be read as a resolved representation of the modular transform: an irreducible atypical character is expressed in the continuous $\{b', j'\}$ character family, while irreducible atypical target contributions are not separated explicitly. The atypicality loci appear as poles of the kernel described above.

Within the Fresnel-Abel prescription, the result gives the S -transform in the $\{b, j\}$ character basis. It does not define a finite S -matrix on the irreducible atypical characters themselves. Moreover, modular invariance of a bulk partition function requires a specified left-right spectrum and measure and invariance under both S and T , including the appropriate spin-structure completion; see, for example, Ref. [SS07, arXiv, Sec. 4.3]. For the coset calculation below, the typical and resolved atypical denominator terms can therefore be treated in the same $\{b, j\}$ character basis.

15. Modular transformation of $\widehat{\mathfrak{psu}}(1,1|2)_k$ atypical characters

The branching functions of the irreducible atypical $\widehat{\mathfrak{psu}}(1,1|2)_k$ characters were derived in Sec. 10. For their modular transformation we use the formal branch resolution in Eq. (33), which expresses each irreducible atypical character in the same discrete-series character family used in the modular transformation of typical characters. The atypical character functions may alternatively be expressed in terms of derivatives of higher-level Appell functions [GQS07, arXiv, Sec. 3.3.2], whose general modular transformation properties are known [STT05]. Here we instead use the resolution-based representation throughout.

In the resolution in Eq. (33), the n -th term has $j_1 = -(j + 1 + \frac{n}{2})$ and $j_2 = j + \frac{n}{2}$. Applying the discrete-series modular transformation in Eq. (67) to this term, the spectral-flow dependence enters through the factor $e^{2\pi i(j_2+s j_1)w}$. For the $s = +1$ branch its exponent is $j_2 + j_1 = -1$, whereas for the $s = -1$ branch it is $j_2 - j_1 = 2j + n + 1 \in \mathbb{Z}$. Hence in both cases $e^{2\pi i(j_2+s j_1)w} = 1$, so the kernel obtained after summing the resolution is independent of the spectral-flow label w . Thus, termwise application of Eq. (67) to the resolution gives

$$\begin{aligned} \chi_{[j]_s}^{(k)} \left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau} \right) &= \frac{2}{k} e^{\frac{\pi i k}{2\tau}(y^2-x^2)} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \sum_{n=1}^{\infty} n \\ &\times \sin \left(\frac{\pi}{k} (2j + n + 1)(2j'_2 + 1) \right) e^{-\frac{\pi i}{k} (2j+n+1)(2j'_1+1)} \\ &\times \gamma_-^w [\chi_{[(s,j'_1),j'_2]}^{(k)}(x, y, \tau)]. \end{aligned} \quad (77)$$

The positive overall sign results from the cancellation of the minus sign in the atypical resolution with that in Eq. (67). Using

$$\sin \left(\frac{\pi}{k} (2j + n + 1)(2j'_2 + 1) \right) = \frac{1}{2i} \sum_{a=\pm 1} a e^{\frac{\pi i}{k} a(2j+n+1)(2j'_2+1)},$$

and defining $X_a := a(2j'_2 + 1) - (2j'_1 + 1)$, the part of the integrand depending on n and the representation labels becomes

$$-\frac{i}{k} \sum_{a=\pm 1} a \sum_{n=1}^{\infty} n e^{\frac{\pi i}{k} (2j+n+1)X_a}.$$

The additional factor n inherited from the atypical resolution is the essential difference from the $\widehat{\mathfrak{su}}(1|2)_k$ case: the resulting geometric sum is of the form $\sum_{n \geq 1} n q^n$ and therefore produces a squared denominator. As in the $\widehat{\mathfrak{su}}(1|2)_k$ case, the series is defined by Abel regularization. Introducing $e^{-\varepsilon n}$ with $\varepsilon > 0$ and using

$$\sum_{n=1}^{\infty} n r^n = \frac{r}{(1-r)^2}, \quad |r| < 1,$$

we obtain the generalized S -kernel

$$\mathcal{S}_{[(s,j'_1),j'_2]}^{\text{F},([j]_s,k)} := -\frac{i}{k} \lim_{\varepsilon \rightarrow 0^+} \sum_{a=\pm 1} a e^{\frac{\pi i}{k} (2j+1)X_a} \frac{e^{-\varepsilon + \frac{\pi i}{k} X_a}}{(1 - e^{-\varepsilon + \frac{\pi i}{k} X_a})^2}. \quad (78)$$

Here the Abel prescription is the analogue of that used in Eq. (73). The superscript F records the Fresnel prescription inherited from the unfolded non-compact modular transformation; after decomposition into spectral-flow sectors, the j'_1 -integration is over the physical interval, with the singular kernel understood in the Abel boundary-value sense specified above. Away from the poles, using

$$\frac{e^{\frac{\pi i}{k} X}}{(1 - e^{\frac{\pi i}{k} X})^2} = -\frac{1}{4 \sin^2(\frac{\pi X}{2k})},$$

the kernel can equivalently be written as

$$\mathcal{S}_{[(s,j'_1),j'_2]}^{\text{F},([j]_s,k)} = \frac{i}{4k} \sum_{a=\pm 1} a \frac{e^{\frac{\pi i}{k}(2j+1)X_a}}{\sin^2\left(\frac{\pi}{2k}X_a\right)}. \quad (79)$$

The modular transformation can therefore be written as

$$\chi_{[j]_s}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) = e^{\frac{\pi i k}{2\tau}(y^2-x^2)} \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \mathcal{S}_{[(s,j'_1),j'_2]}^{\text{F},([j]_s,k)} \gamma_-^w [\chi_{[(s,j'_1),j'_2]}^{(k)}(x, y, \tau)]. \quad (80)$$

The meromorphic form of the kernel has double poles whenever $X_a = 2km$, $m \in \mathbb{Z}$, since the denominator $\sin^2(\frac{\pi X_a}{2k})$ then vanishes quadratically. With $X_a = a(2j'_2 + 1) - (2j'_1 + 1)$, the two possibilities are $a = +1 : j'_1 = j'_2 - km$ and $a = -1 : j'_1 = -j'_2 - 1 - km$. In the physical integration range $-\frac{k+1}{2} < j'_1 < -\frac{1}{2}$ with $0 \leq j'_2 \leq \frac{k-2}{2}$, only the solution $a = -1$, $m = 0$ can occur. It gives $j'_1 + j'_2 + 1 = 0$, which is precisely the atypicality condition for the unflowed discrete series. Thus the poles located on the physical j'_1 contour occur exactly at the reducible atypical labels. The remaining solutions belong to the periodically continued pole lattice outside the physical strip.

The vacuum follows from the formal continuation $j \rightarrow -\frac{1}{2}$: by Eq. (27), $\chi_{[0]}^{(k)} = -\chi_{[-1/2]_s}^{(k)}$ for either branch $s = \pm 1$. The additional minus sign therefore gives

$$\mathcal{S}_{[(s,j'_1),j'_2]}^{\text{F},([0],k)} = -\frac{i}{4k} \sum_{a=\pm 1} a \frac{1}{\sin^2\left[\frac{\pi}{2k}(a(2j'_2 + 1) - (2j'_1 + 1))\right]}. \quad (81)$$

As explained below Eq. (27), the two choices $s = \pm 1$ correspond to the two Laurent expansions of the same meromorphic vacuum character and are not to be added. For $k = 2$ the compact sum contains only $j'_2 = 0$, and Eq. (81) reduces to

$$\mathcal{S}_{[(s,j'_1),0]}^{\text{F},([0],2)} = -\frac{i}{8} \left(\frac{1}{\sin^2\left(\frac{\pi j'_1}{2}\right)} - \frac{1}{\cos^2\left(\frac{\pi j'_1}{2}\right)} \right). \quad (82)$$

Eq. (80) therefore gives, within the Fresnel-Abel prescription, the modular transform of an irreducible atypical character as a sum over spectral-flowed discrete-series characters. The corresponding kernel is meromorphic in j'_1 and has double poles at the atypicality loci identified above. The irreducible atypical target contributions are therefore not separated explicitly; instead, the kernel has poles at their atypicality labels. It does not define a finite S -matrix on the irreducible atypical characters themselves.

16. Modular transformation of coset characters

The modular transformation of the coset branching functions follows from the character decomposition itself. In the present coset, the denominator algebra $\mathfrak{su}(1|2)$ is the stabilizer algebra of the homogeneous-space geometry. Thus the same restriction from $\mathfrak{psu}(1,1|2)$ to $\mathfrak{su}(1|2)$ that appears geometrically also underlies the character branching. The modular transformation of the branching functions is then fixed by requiring this decomposition to be compatible with the

modular transformations of both numerator and denominator characters. This compatibility is expressed by the intertwining relation derived below.

For a $\widehat{\mathfrak{psu}}(1,1|2)_k$ numerator character labelled by Λ we write schematically

$$\chi_\Lambda^{(k)}(x, y, \tau) = \sum_\lambda \Psi_\lambda^{(\Lambda, k)}(\tau) \chi_\lambda^{(k)}(x, y, \tau),$$

where λ labels the corresponding $\widehat{\mathfrak{su}}(1|2)_k$ denominator characters. Depending on the decomposition, $\sum_\lambda$ denotes ordinary sums, continuous integrals, or combinations thereof. The Fresnel prescription is that of Eq. (64), while the Abel boundary values for atypical characters are those fixed in Secs. 14 and 15.

The numerator label Λ ranges over the $\widehat{\mathfrak{psu}}(1,1|2)_k$ character families considered above, including the discrete series, the principal continuous series and their spectral-flow sectors, as well as the atypical representations. The denominator label λ will be represented below by the $\widehat{\mathfrak{su}}(1|2)_k$ Kac characters $\chi_{\{p/2, t/2\}}^{(k)}$, with $p \in \mathbb{R}$ and $t = 1, \dots, k$. For the unflowed numerator characters, the denominator charges occurring in the branching decomposition lie in an allowed set $L_{\Lambda, t} \subset \mathbb{R}$, which will be given explicitly below. At the denominator reducibility loci $p = \pm t + 2r(k+1)$, $r \in \mathbb{Z}$, these Kac characters are reducible and atypical; away from these loci they are irreducible and typical.

Irreducible atypical denominator characters that occur separately in the original branching decomposition will be resolved into the same Kac-character family before applying the denominator modular kernel. This resolution is not required: one could instead use their direct Appell-Mordell transformation and keep the irreducible atypical terms explicit. We use the resolution here because it expresses all denominator contributions in the same character family, so that their branching coefficients can be compared directly.

Let the generalized modular kernels of numerator and denominator be denoted by $\mathcal{S}_{\Lambda\Lambda'}^{\text{PSU}}$ and $\mathcal{S}_{\lambda\lambda'}^{\text{SU}}$, respectively, so that schematically

$$\chi_\Lambda^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) = J_k(x, y, \tau) \sum_{\Lambda'} \mathcal{S}_{\Lambda\Lambda'}^{\text{PSU}} \chi_{\Lambda'}^{(k)}(x, y, \tau),$$

and

$$\chi_\lambda^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) = J_k(x, y, \tau) \sum_{\lambda'} \mathcal{S}_{\lambda\lambda'}^{\text{SU}} \chi_{\lambda'}^{(k)}(x, y, \tau),$$

with the common Jacobi automorphy factor

$$J_k(x, y, \tau) := e^{\frac{\pi i k}{2\tau}(y^2 - x^2)}. \quad (83)$$

Inserting these transformations into the branching decomposition, the common factor J_k cancels. Comparing the coefficients of $\chi_{\lambda'}^{(k)}$ gives the generalized intertwining relation

$$\sum_\lambda \Psi_\lambda^{(\Lambda, k)}\left(-\frac{1}{\tau}\right) \mathcal{S}_{\lambda\lambda'}^{\text{SU}} = \sum_{\Lambda'} \mathcal{S}_{\Lambda\Lambda'}^{\text{PSU}} \Psi_{\Lambda'}^{(\Lambda', k)}(\tau). \quad (84)$$

The left-hand side is obtained by first branching the transformed numerator character and then transforming the denominator characters, whereas the right-hand side is obtained by first transforming the numerator character and then applying the branching decomposition. All sums and

integrals are understood with the prescriptions appropriate to the corresponding character families.

Although explicit expressions for the branching functions were derived in Secs. 8–10, they need not be transformed term by term. These expressions determine the coset characters themselves, including their allowed denominator charges and the affine grades at which the individual denominator characters occur. Their modular transformation, however, follows more directly from the full character decomposition through Eq. (84). This avoids reconstructing the diagonal branching structure from the explicit q -series after the modular transformation.

Below we derive the inverse of the denominator Fourier-sine kernel on the continuously labelled Kac-character family explicitly, using the Fourier delta identity and the finite sine orthogonality relation. Away from the denominator reducibility loci this gives the ordinary inverse of the typical character transform. At reducible Kac labels the same regular Kac-character kernel continues to apply; only the representation-theoretic interpretation of the corresponding character changes, as specified below. In either case, whenever this inverse character kernel is applicable, Eq. (84) may be written symbolically as

$$\Psi\left(-\frac{1}{\tau}\right) = \mathcal{S}^{\text{PSU}} \Psi(\tau) (\mathcal{S}^{\text{SU}})^{-1}.$$

For branching decompositions containing irreducible atypical $\widehat{\mathfrak{su}}(1|2)_k$ denominator characters, the preceding symbolic formula is understood only after resolving those characters into the common Kac-character family. As shown in Sec. 14, the irreducible atypical characters themselves transform through a generalized meromorphic kernel into the Kac-character family, rather than through a finite matrix among the irreducible atypical characters. No finite invertible $\mathcal{S}$ -matrix among the irreducible atypical denominator characters is therefore required here.

To apply the intertwining relation with the continuous Fourier kernel of the denominator theory, we now rewrite the discrete denominator charge sums as distributions on the real p -axis. For fixed numerator label Λ and $t = 1, \dots, k$, let $L_{\Lambda,t} \subset \mathbb{R}$ denote the set of allowed values of the denominator charge p in the corresponding branching decomposition. We define

$$\widetilde{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{(\Lambda,k)}(\tau) := \sum_{m \in L_{\Lambda,t}} \delta(p - m) \Psi_{\{\frac{m}{2}, \frac{t}{2}\}}^{(\Lambda,k)}(\tau). \quad (85)$$

Thus $\widetilde{\Psi}$ introduces no new branching data: it represents the original branching functions, defined at their allowed discrete charges, as a distribution in the continuous variable p . The corresponding part of the character decomposition can then be written as

$$\chi_{\Lambda}^{(k)}(x, y, \tau) = \sum_{t=1}^k \int_{\mathbb{R}} dp \, \widetilde{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{(\Lambda,k)}(\tau) \chi_{\{\frac{p}{2}, \frac{t}{2}\}}^{(k)}(x, y, \tau). \quad (86)$$

For branching decompositions containing separate irreducible atypical denominator characters, Eq. (86) will apply after these terms have been resolved into the same Kac-character family, as described below.

For a finite-ground-state numerator, the branching condition in Eq. (53) gives

$$L_{[j_1, j_2], t} = \{m \in \mathbb{Z} \mid m + t \equiv 2(j_1 + j_2) \bmod 2\}.$$

Similarly, from the discrete-series decomposition in Eq. (56),

$$L_{[(s,j_1),j_2],t} = -2sj_1 + t - 2j_2 + 2\mathbb{Z}.$$

To express Eq. (69) in the notation used for the branching distributions above, set $K := k + 1$ and identify $b = p/2$, $b' = p'/2$, $j = t/2$, and $j' = t'/2$. Since $db' = \frac{1}{2}dp'$, the transformation becomes

$$\chi_{\{\frac{p}{2}, \frac{t}{2}\}}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) = J_k(x, y, \tau) \int_{\mathbb{R}} dp' \sum_{t'=1}^k \mathcal{S}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{\text{F}, (\{\frac{p}{2}, \frac{t}{2}\}, k)} \chi_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{(k)}(x, y, \tau), \quad (87)$$

where

$$\mathcal{S}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{\text{F}, (\{\frac{p}{2}, \frac{t}{2}\}, k)} = -\frac{i}{K} e^{\frac{\pi i}{K} p p'} \sin\left(\frac{\pi t t'}{K}\right). \quad (88)$$

As above, the superscript F records the Fresnel prescription for the continuous charge integration. On this continuously labelled Kac-character family, the inverse kernel is

$$(\mathcal{S}^{-1})_{\{\frac{p}{2}, \frac{t}{2}\}}^{\text{F}, (\{\frac{p'}{2}, \frac{t'}{2}\}, k)} = \frac{i}{K} e^{-\frac{\pi i}{K} p p'} \sin\left(\frac{\pi t t'}{K}\right). \quad (89)$$

Indeed, composing it with Eq. (88) reduces to the standard distributional Fourier identity and finite sine orthogonality relation,

$$\int_{\mathbb{R}} dp' e^{\frac{\pi i}{K}(p-q)p'} = 2K \delta(p-q), \quad \sum_{t'=1}^k \sin\left(\frac{\pi t t'}{K}\right) \sin\left(\frac{\pi u t'}{K}\right) = \frac{K}{2} \delta_{t,u},$$

whose normalization factors cancel exactly.

We first use this inverse away from the denominator reducibility loci $p = \pm t + 2rK$, $r \in \mathbb{Z}$. For a finite-ground-state numerator, the branching condition $m+t \equiv 2(j_1+j_2) \pmod{2}$ shows that the charge lattice $L_{[j_1, j_2], t}$ intersects these loci precisely when $j_1+j_2 \in \mathbb{Z}$, since $m+t$ is even for $m = \pm t + 2rK$. The Fourier-sine character kernel itself is regular at these labels and extends to them by the same Kac-character formula. What changes there is only the representation-theoretic interpretation: the Kac module becomes reducible.

We first consider finite-ground-state numerators. By Eq. (32), their characters are formally obtained by adding the two continued discrete-series branches. It is therefore sufficient to determine the modular transformation of the branching distribution for a single branch $[(s, j_1), j_2]$ and subsequently sum over $s = \pm 1$. For this purpose it is convenient to use the unfolded form of the numerator transformation in Eq. (67), in which the transformed non-compact label j'_1 runs over $\mathbb{R}$ before the integral is decomposed into the physical discrete-series interval and its spectral-flow translates. Combining the corresponding $\widehat{\mathfrak{psu}}(1,1|2)_k$ kernel with the inverse denominator kernel in Eq. (89) gives

$$\begin{aligned} \widetilde{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([(s, j_1), j_2], k)}\left(-\frac{1}{\tau}\right) &= -\frac{2i}{kK} \int_{\mathbb{R}} dj'_1 \sum_{2j'_2=0}^{k-2} e^{\frac{\pi i}{k}(2j_1+1)(2j'_1+1)} \sin\left(\frac{\pi}{k}(2j_2+1)(2j'_2+1)\right) \\ &\times \int_{\mathbb{R}} dp' \sum_{t'=1}^k e^{-\frac{\pi i}{K} p p'} \sin\left(\frac{\pi t t'}{K}\right) \widetilde{\Psi}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{([(s, j'_1), j'_2], k)}(\tau). \end{aligned} \quad (90)$$

The first line contains the numerator modular kernel, while the second line contains the inverse denominator Fourier-sine kernel. The overall factor $-2i/(kK)$ is correspondingly the product of the numerator factor $-2/k$ and the denominator inverse factor i/K .

Eq. (90) is thus the intertwining relation for a single discrete-series branch. For $j_1 = 0, \frac{1}{2}, \dots, \frac{k+2}{2}$, $j_2 = 0, \frac{1}{2}, \dots, \frac{k-2}{2}$, and $j_1 \neq j_2$, Eq. (32) then implies that the finite-ground-state branching distribution is obtained by summing Eq. (90) over the two formally continued branches $s = \pm 1$. The unfolded j'_1 -integral may subsequently be decomposed into the physical discrete-series interval and its spectral-flow translates.

We next consider a principal continuous numerator with $j_1 \in -\frac{1}{2} + i\mathbb{R}$, $0 \leq \alpha < 1$, and $j_2 = 0, \frac{1}{2}, \dots, \frac{k-2}{2}$. From the branching decomposition derived above, its denominator charge lattice is $L_{[(j_1, \alpha), j_2], t} = 2\alpha + t - 2j_2 + 2\mathbb{Z}$. Since the modular transformation in Eq. (68) generates the complete set of spectral-flow sectors, we define the corresponding flowed branching distributions implicitly by

$$\gamma_-^w[\chi_{[(j_1, \alpha), j_2]}^{(k)}](x, y, \tau) = \sum_{t=1}^k \int_{\mathbb{R}} dp \, \widetilde{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([(j_1, \alpha), j_2]_w, k)}(\tau) \chi_{\{\frac{p}{2}, \frac{t}{2}\}}^{(k)}(x, y, \tau).$$

The spectral-flow label is kept explicit, since a flowed principal continuous character cannot in general be absorbed into shifted unflowed representation labels. Combining Eq. (68) with the inverse denominator kernel in Eq. (89) gives

$$\begin{aligned} \widetilde{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([(j_1, \alpha), j_2]_w, k)}\left(-\frac{1}{\tau}\right) &= \frac{2i}{kK} \int_{-\frac{1}{2}+i\mathbb{R}} dj'_1 \int_0^1 d\alpha' \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \\ &\times e^{\frac{\pi i}{k}(2j_1+1)(2j'_1+1)} \sin\left(\frac{\pi}{k}(2j_2+1)(2j'_2+1)\right) e^{2\pi i(j_2-\alpha)w} \\ &\times \int_{\mathbb{R}}^F dp' \sum_{t'=1}^k e^{-\frac{\pi i}{K}pp'} \sin\left(\frac{\pi tt'}{K}\right) \widetilde{\Psi}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{([(j'_1, \alpha'), j'_2]_w, k)}(\tau). \end{aligned} \quad (91)$$

Although the principal continuous numerator is always typical, its denominator charge lattice can intersect the reducibility loci $p = \pm t + 2rK$, $r \in \mathbb{Z}$. Using $L_{[(j_1, \alpha), j_2], t} = 2\alpha + t - 2j_2 + 2\mathbb{Z}$, this occurs precisely when $\alpha - j_2 \in \mathbb{Z}$. Hence, for $0 \leq \alpha < 1$, the exceptional values are $\alpha = 0$ for integral j_2 and $\alpha = \frac{1}{2}$ for half-integral j_2 . At these labels the denominator Kac character is reducible, but the Fourier-sine Kac-character kernel remains regular, so Eq. (91) continues to apply without modification. No resolution is needed merely because a Kac character becomes reducible at these labels. The resolution enters only for the atypical numerator characters discussed below, whose original branching decomposition contains separate irreducible atypical denominator characters.

We finally turn to these atypical numerator characters. Their branching decomposition, given by Eqs. (59) and (60), contains both irreducible atypical and irreducible typical denominator characters. For the modular calculation, we resolve the atypical terms into the same continuously labelled Kac-character family used above, so that all denominator contributions are expressed in a common character basis.

For each non-vacuum irreducible atypical denominator character $\chi_{\{j'\}_b}^{(k)}$, $b = \pm 1$, occurring in this decomposition, we rewrite its formal Kac resolution in the continuously labelled Kac-

character basis used above. For each term of Eq. (71) whose compact label does not lie on an affine Weyl wall, we fold

$$2j' + n = d_n t_n + 2r_n K, \quad d_n = \pm 1, \quad r_n \in \mathbb{Z}, \quad t_n = 1, \dots, k.$$

The affine Weyl continuation of the compact factor then replaces the second label $j' + n/2$ by $t_n/2$ and contributes the sign d_n , while the charge label $b(j' + n/2)$ remains unchanged. If $2j' + n \in K\mathbb{Z}$, the compact label lies on an affine Weyl wall and the corresponding character vanishes. These terms therefore do not contribute to the resolution. The remaining discrete resolution can be represented in the common (p, t) parametrization by

$$R_{p,t}^{(\{j'\}_b, k)} = - \sum_{\substack{n \geq 1 \\ 2j' + n \notin K\mathbb{Z}}} d_n \delta_{t, t_n} \delta(p - b(2j' + n)),$$

so that

$$\sum_{t=1}^k \int_{\mathbb{R}} dp R_{p,t}^{(\{j'\}_b, k)} \chi_{\{\frac{p}{2}, \frac{t}{2}\}}^{(k)} = - \sum_{\substack{n \geq 1 \\ 2j' + n \notin K\mathbb{Z}}} \chi_{\{b(j' + \frac{n}{2}), j' + \frac{n}{2}\}}^{(k)} = - \sum_{n=1}^{\infty} \chi_{\{b(j' + \frac{n}{2}), j' + \frac{n}{2}\}}^{(k)} = \chi_{\{j'\}_b}^{(k)},$$

where the second equality simply reinstates the vanishing Weyl-wall terms, and the last equality is Eq. (71). This form allows the resolved atypical contributions to enter the same modular intertwining relation as the generic Kac characters. For the denominator vacuum $\{0\}$, the same construction applies with $j' = 0$ after choosing one of the two formal branches $s = \pm 1$ in Eq. (71). These give equivalent meromorphic resolutions of the same vacuum character and must not be added. We fix one of the two branches once and for all and denote the corresponding resolution distribution by $R_{p,t}^{(\{0\}, k)}$ throughout the following. The two choices need not give identical coefficient distributions, although they reconstruct the same vacuum character. Using Eqs. (59) and (60), we can now combine the typical branching distribution with the resolved atypical contributions. We define

$$\widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}(\tau) := \widetilde{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}(\tau) + \Psi_{\{0\}}^{([j]_s, k)}(\tau) R_{p,t}^{(\{0\}, k)} + \sum_{n \in \mathbb{N}} \sum_{b=\pm 1} \Psi_{\{\frac{n}{2}\}_b}^{([j]_s, k)}(\tau) R_{p,t}^{(\{\frac{n}{2}\}_b, k)}. \quad (92)$$

Here $\widehat{\Psi}$ denotes the resolved branching distribution in the common Kac-character basis. Its first term, $\widetilde{\Psi}_{\{p/2, t/2\}}^{([j]_s, k)}$, is the distributional embedding of the branching functions for typical denominator characters defined in Eq. (85). The second term is the contribution of the denominator vacuum, while the sum in the last term contains the remaining irreducible atypical denominator contributions. In the latter two terms the original branching functions $\Psi_{\{0\}}$ and $\Psi_{\{n/2\}_b}$ remain unchanged; their associated irreducible atypical characters are replaced by the resolution distributions $R_{p,t}$.

For the transformed numerator characters appearing in Eq. (80), we define the branching distributions of the discrete-series spectral-flow sectors implicitly by

$$\gamma_-^w [\chi_{[(s, j'_1), j'_2]}^{(k)}](x, y, \tau) = \sum_{t'=1}^k \int_{\mathbb{R}} dp' \widetilde{\Psi}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{([l(s, j'_1), j'_2]_w, k)}(\tau) \chi_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{(k)}(x, y, \tau).$$

Thus the spectral-flow label is retained explicitly on the numerator branching distribution. We now evaluate the modular transform of the same atypical numerator character in two ways that must agree. In the first route, we start from its resolved denominator decomposition, with the resolved branching distribution $\widehat{\Psi}$ defined in Eq. (92), and transform the denominator Kac characters. In the second route, we first apply the numerator modular transformation in Eq. (80). This produces spectral-flowed discrete-series numerator characters, whose denominator branching distributions are the $\widetilde{\Psi}$ just defined.

In the first route, the resolved decomposition of the untransformed atypical numerator character is

$$\chi_{[j]_s}^{(k)}(x, y, \tau) = \sum_{t=1}^k \int_{\mathbb{R}} dp \widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}(\tau) \chi_{\{\frac{p}{2}, \frac{t}{2}\}}^{(k)}(x, y, \tau).$$

After replacing τ by $-1/\tau$, the coefficient becomes $\widehat{\Psi}(-1/\tau)$. Applying the denominator transformation Eq. (87) to the Kac characters then gives

$$\begin{aligned} \chi_{[j]_s}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) &= J_k(x, y, \tau) \int_{\mathbb{R}} dp' \sum_{t'=1}^k \\ &\times \left[\sum_{t=1}^k \int_{\mathbb{R}} dp \widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}\left(-\frac{1}{\tau}\right) \mathcal{S}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{F, (\{\frac{p}{2}, \frac{t}{2}\}, k)} \right] \chi_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{(k)}(x, y, \tau). \end{aligned}$$

Here $\widehat{\Psi}$ contains the three contributions defined above: the embedded typical branching distribution, the resolved denominator vacuum contribution, and the remaining resolved irreducible atypical denominator contributions.

In the second route, Eq. (80) first transforms the atypical numerator character into a sum over spectral-flowed discrete-series numerator characters. Branching each of these target characters with the distributions defined above gives

$$\begin{aligned} \chi_{[j]_s}^{(k)}\left(\frac{x}{\tau}, \frac{y}{\tau}, -\frac{1}{\tau}\right) &= J_k(x, y, \tau) \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \mathcal{S}_{[(s, j'_1), j'_2]}^{F, ([j]_s, k)} \\ &\times \int_{\mathbb{R}} dp' \sum_{t'=1}^k \widetilde{\Psi}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{([(s, j'_1), j'_2]_w, k)}(\tau) \chi_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{(k)}(x, y, \tau). \end{aligned}$$

Comparing the coefficients of $\chi_{\{p'/2, t'/2\}}^{(k)}$ yields

$$\sum_{t=1}^k \int_{\mathbb{R}} dp \widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}\left(-\frac{1}{\tau}\right) \mathcal{S}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{F, (\{\frac{p}{2}, \frac{t}{2}\}, k)} = \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \mathcal{S}_{[(s, j'_1), j'_2]}^{F, ([j]_s, k)} \widetilde{\Psi}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{([(s, j'_1), j'_2]_w, k)}(\tau).$$

Applying the inverse denominator kernel in Eq. (89) to this relation gives

$$\begin{aligned} \widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}\left(-\frac{1}{\tau}\right) &= \int_{-\frac{k+1}{2}}^{-\frac{1}{2}} dj'_1 \sum_{2j'_2=0}^{k-2} \sum_{w \in \mathbb{Z}} \mathcal{S}_{[(s, j'_1), j'_2]}^{F, ([j]_s, k)} \\ &\times \int_{\mathbb{R}} dp' \sum_{t'=1}^k (\mathcal{S}^{-1})_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{F, (\{\frac{p'}{2}, \frac{t'}{2}\}, k)} \widetilde{\Psi}_{\{\frac{p'}{2}, \frac{t'}{2}\}}^{([(s, j'_1), j'_2]_w, k)}(\tau). \end{aligned} \tag{93}$$

Here the first kernel is the generalized numerator kernel defined in Eq. (78), while the second is the inverse denominator kernel in Eq. (89). The spectral-flow label w appears only on the target branching distributions; the numerator kernel itself is independent of w , as shown above Eq. (77). The double poles of the numerator kernel are understood with the Abel prescription used in Eq. (78).

For the finite-ground-state atypicals with $j = \frac{1}{2}, 1, \dots, \frac{k-2}{2}$, no additional modular calculation is required. By the character relation Eq. (26) and linearity of the resolved denominator decomposition,

$$\widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j], k)} = - \sum_{s=\pm 1} \widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([j]_s, k)}. \quad (94)$$

Its modular transformation is therefore obtained directly by applying Eq. (93) to the two branches and summing them with the same overall minus sign.

The numerator vacuum is exceptional. By Eq. (27), it is obtained from either one of the two formal branch continuations, with a single branch chosen throughout. Consequently,

$$\widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([0], k)} = -\widehat{\Psi}_{\{\frac{p}{2}, \frac{t}{2}\}}^{([\frac{1}{2}]_s, k)}, \quad (95)$$

with s fixed by the chosen branch. Its modular transformation follows directly from Eq. (93) by the formal continuation $j \rightarrow -\frac{1}{2}$ together with the minus sign in Eq. (95). Equivalently, the numerator kernel is the vacuum kernel $\mathcal{S}_{[(s, j'), j'_2]}^{F, ([0], k)}$ defined in Eq. (81).

With these kernels and resolved branching distributions, the modular S -transformation of the coset branching distributions is obtained by intertwining the generalized numerator transformations with the inverse denominator transformation. For typical numerator characters, this acts directly on the continuously embedded branching distributions. At reducible denominator labels the same Fourier-sine Kac-character kernel remains regular; only the corresponding Kac module becomes reducible. When irreducible atypical denominator characters occur explicitly in the branching decomposition, they are first resolved into the Kac-character basis. The resulting resolved branching distributions $\widehat{\Psi}$ then transform by the same intertwining procedure, with the generalized atypical numerator kernels derived in Sec. 15. Their pole prescription is inherited from that numerator transformation and does not require a finite S -matrix among irreducible atypical characters. Thus typical and atypical coset branching distributions are treated in the same denominator Kac-character basis, while spectral-flow sectors remain explicit in the transformed numerator characters.

17. Conclusion

In this work we studied the character decomposition and modular properties of the WZW coset

$$\frac{\text{PSU}(1,1|2)}{\text{SU}(1|2)}.$$

As discussed in Sec. 2, this model is the non-compact, hyperbolic counterpart of the compact $\text{PSU}(2|2)/\text{SU}(1|2)$ coset associated with $\mathbb{CP}^{1|2}$. The denominator algebra $\mathfrak{su}(1|2)$ is the stabilizer algebra of the corresponding homogeneous superspace. At the level of affine current algebras, the coset characters therefore arise from the restriction of $\widehat{\mathfrak{psu}}(1,1|2)_k$ representations to

$\widehat{\mathfrak{su}}(1|2)_k$. Starting from the finite-dimensional branching rules, we determined the corresponding affine branching functions and their modular transformation properties.

For typical representations with finite-dimensional ground states, the affine branching derived in Sec. 8 factorizes into a left diagonal branching involving finite-Weyl negative-level $\widehat{\mathfrak{sl}}(2)$ characters and a compact $\widehat{\mathfrak{su}}(2)$ diagonal branching. The lowest-grade terms of the resulting affine decomposition reproduce the corresponding finite-dimensional typical branching rule, including its signed multiplicities. Using the fixed-charge decompositions of the negative-level $\widehat{\mathfrak{sl}}(2)$ characters derived in Sec. 7, Sec. 9 extends the construction to representations whose ground states belong to the discrete and principal continuous series of the non-compact $\mathfrak{sl}(2, \mathbb{R})$ factor. The atypical analysis in Sec. 10 starts instead from the atypical discrete-series branches $[j]_s$. Their branching separates the irreducible atypical denominator characters occurring explicitly in the decomposition from the complementary typical contributions, and the two branches are then combined to reconstruct the finite-ground-state atypical characters. The exceptional vacuum representation is obtained from a single continued branch rather than from the sum of the two branch continuations. The direct low-order expansion of the numerator vacuum character in Sec. 5 provides an independent check of the resulting denominator characters and coefficients.

Starting from the known character formulas and spectral-flow structure of $\widehat{\mathfrak{psu}}(1,1|2)_k$, we derived in Sec. 12 the modular transformations of the typical characters while keeping the spectral-flow labels explicit. For the discrete series the spectral flow may alternatively be represented through unfolded labels, whereas the explicit label provides a common description that also applies to the principal continuous series. Within the meromorphic/Fresnel prescription used here, the complete spectral-flow families of both series close under the S -transformation. Using the auxiliary bulk state space $\widehat{\mathcal{H}}_0$, Sec. 13 then tests the modular closure of its typical part. The corresponding diagonal character sums transform covariantly with the expected Jacobi factor and are formally S -invariant at vanishing chemical potentials. The T -transformation is diagonal, and its conformal phases cancel between the holomorphic and antiholomorphic characters in the diagonal pairing. This establishes modular closure of the typical part of the auxiliary bulk state space within the stated prescription. As discussed in Ref. [GQS07, arXiv, Sec. 6], the interacting numerator theory has in addition a non-factorizing indecomposable structure in its atypical bulk sector. The modular-closure test performed here therefore does not constitute a construction of the complete interacting torus partition function.

The atypical characters were treated through formal resolutions in Secs. 14 and 15. For $\widehat{\mathfrak{su}}(1|2)_k$, an irreducible atypical character is resolved into the same continuously labelled Kac-character family used for the generic denominator characters. Applying the Kac-character modular transformation term by term gives a continuous distribution in this basis, with a meromorphic kernel whose poles occur at the atypicality loci. The analogous resolution for $\widehat{\mathfrak{psu}}(1,1|2)_k$ expresses the irreducible atypical characters in the discrete-series character family and, after modular transformation, gives a sum over complete spectral-flow orbits with a meromorphic kernel possessing double poles at the corresponding atypicality loci. These formulas are understood with the Abel and Fresnel prescriptions specified in the text. This resolution-based treatment is an alternative to the Appell-Mordell description of the denominator atypicals and the derivative higher-level Appell description of the numerator atypicals mentioned in Secs. 14 and 15. The atypical modular transformations are therefore not described here by finite S -matrices among irreducible atypical characters. Instead, the denominator and numerator atypical characters are

treated through resolutions into the Kac-character and discrete-series character families, respectively, on which the corresponding modular kernels act.

Finally, Sec. 16 derives the modular transformation of the coset branching functions directly from the full numerator-denominator character decomposition. The common Jacobi automorphy factor cancels, and comparison of the denominator characters gives the generalized intertwining relation between the numerator modular kernel, the denominator modular kernel and the branching functions. This avoids transforming the explicit q -series for the branching functions term by term.

A technical step in making this relation explicit is the distributional embedding introduced in Eqs. (85) and (86). The original branching functions are defined only at the discrete denominator charges allowed by the corresponding numerator representation. Replacing these discrete charge sums by weighted delta-function distributions supported on the allowed charge lattices introduces no new branching data, but places them in the continuously labelled Kac-character basis on which the denominator Fourier-sine modular kernel acts. Inserting the definition and performing the charge integral reproduces the original discrete decomposition. The inverse denominator transformation can therefore be applied directly. For branching decompositions containing irreducible atypical denominator characters, their formal resolutions bring these contributions into the same Kac-character basis. The typical and resolved atypical branching distributions can consequently be treated within a single modular intertwining framework, while the spectral-flow structure of the numerator theory remains explicit.

The results obtained here therefore provide the character decomposition and modular transformation data for the non-compact $\text{PSU}(1,1|2)/\text{SU}(1|2)$ coset, where discrete and principal continuous representations, spectral flow and atypicality have to be treated simultaneously. They are ingredients for a modular description of the coset, but do not constitute a construction of a complete interacting modular-invariant bulk theory for the coset. In particular, how the atypical contributions and the non-factorizing indecomposable structures of the interacting numerator theory enter such a complete coset bulk construction lies beyond the scope of the present work.

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